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Maternal mental health liveability shaped by composite residential environmental profiles

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Women are a vulnerable population for mental illnesses, with the perinatal period representing a particularly high-risk susceptible window. Whereas maternal mental health research often narrowly focused on postpartum depression, psychological disturbances during pregnancy received insufficient attention. Here we analyzed a Chinese nationwide 13-year maternity cohort involving 216,442 pregnant women, revealing that the prevalence of prenatal mental distress symptoms peaked at 31.8%. We identified a broad spectrum of risk determinants, spanning socioeconomic disadvantages, behavioral vulnerabilities, medical histories and suboptimal residential environmental exposures. We estimated that 1.61 (95% uncertainty interval: 1.35–1.88) million prenatal mental distress symptoms were attributable to fine particulate matter ($PM_{2.5}$) and heatwave exposure, while green spaces prevented 791.7 (95% uncertainty interval: 648.1–948.9) thousand cases. We further delineated multiple pathways through which urban greening confers mental well-being benefits, including attenuation of pollution-associated stress, reduction of heatwave occurrence, encouragement of outdoor physical activity and intrinsic mental restoration. Our study ultimately provides policy-oriented evidence to inform fertility-supportive and health-promoting urban planning, echoing China's 'three-child policy' and 'Healthy China 2030' blueprint.

Women exhibit 10–25% lifetime susceptibility to major depressive disorder, a rate 2–3 times higher than men, especially during the perinatal period as the critical vulnerable window[1]. Whereas most previous perinatal health studies focused on postpartum depression[2], maternal mental health represents a critical yet under-addressed dimension of population well-being, as prenatal (that is, before delivery, particularly referring to mid-to-late pregnancy) depression has been reported at alarmingly high prevalences, with estimates surpassing 20% in some populations[3–5]. Undetected or untreated prenatal depression can result in poor maternal nutritional status[6], reduced compliance to prenatal healthcare and medical advice[7], inappropriate intake of tobacco, alcohol or psychotropic substances[8,9] and even an elevated risk of suicide[10]. More broadly, prenatal (also known as antenatal) mental distress symptoms (PMDs), encompassing both depressive and anxiety manifestations, are also associated with multiple adverse pregnancy outcomes, including but not limited to spontaneous abortion[11], preterm delivery[12], intrauterine growth restriction[13], low birth weight[14] or small-for-gestational-age[15] at birth.

To date, ever-growing epidemiological evidence has been demonstrating associations between exposure to fine particulate matter (for example, $PM_{2.5}$), extreme heat events exacerbated by climate change and a broad range of health burdens including psychological and psychiatric distress[16–18], whereas residential green spaces are associated with reduced risks of anxiety and depression[19,20]. Population-based associations have, in turn, motivated mechanistic investigations, which suggest that $PM_{2.5}$ may affect mental health through inflammation,

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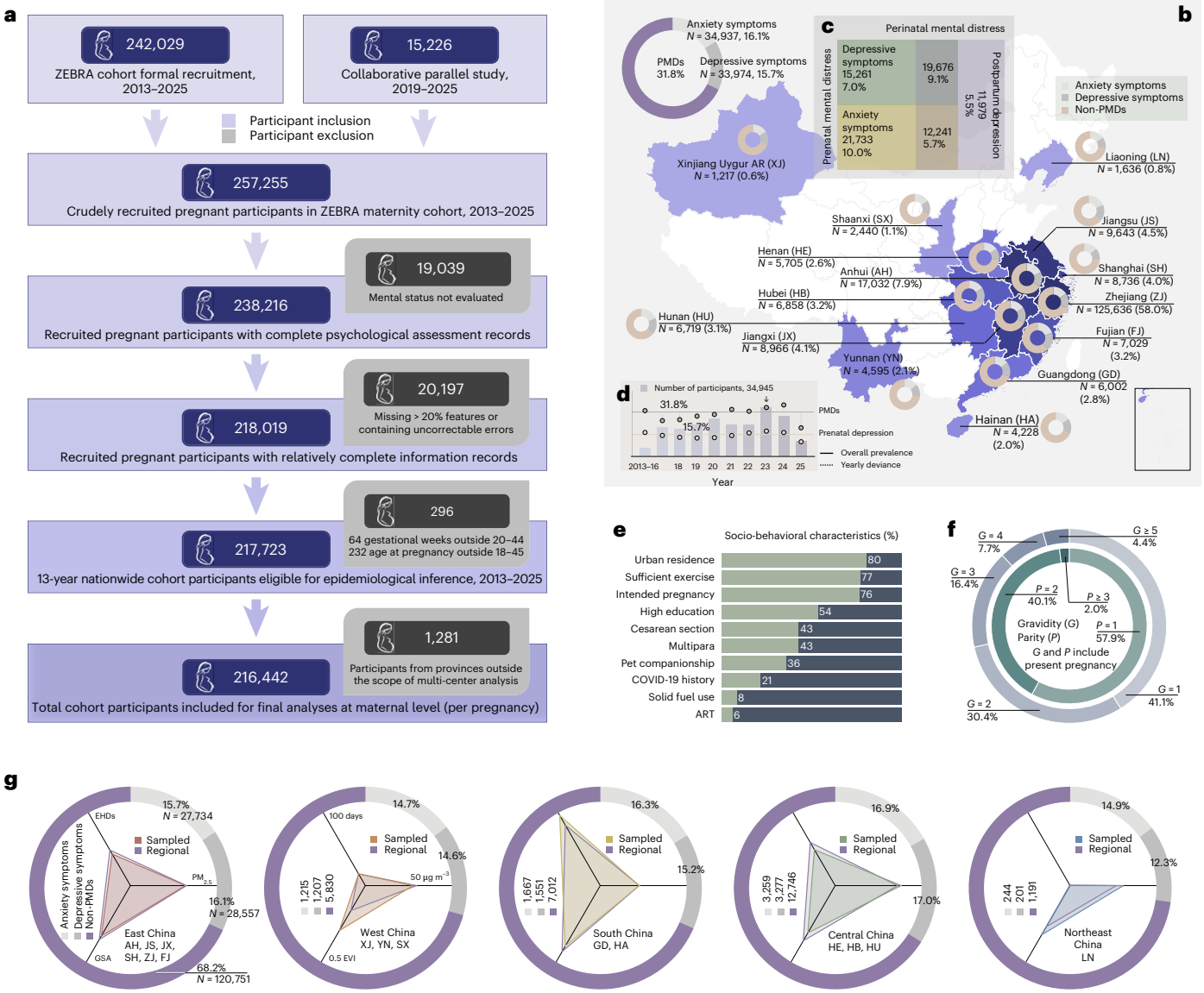


Fig. 1 | ZEBRA maternity cohort profile of recruited participants, 2013–2025. **a**, Recruitment flowchart of ZEBRA and collaborative maternity cohort participants. **b**, Province-clustered geographic distribution of recruited participants across 15 Chinese provinces. Each province is labeled with number of participants and proportion relative to the entire cohort. Donut charts indicate provincial prevalence of prenatal anxiety and depressive symptoms. **c**, Venn diagram for perinatal mental distress symptoms. PMDs include hierarchical subcategories of anxiety and depressive symptoms. Perinatal mental distress symptoms additionally include postnatal depression, a large proportion of which is preceded by PMDs. **d**, Annual number of recruited participants and the corresponding yearly prevalence rates of PMDs and prenatal depression. Overall prevalence estimates were calculated across all 216,442 participants. **e**, Distribution of major socio-behavioral characteristics. 'High education' refers to participants with a college degree or above. 'Multipara' refers to women who have had at least one previous live birth. ART, assisted reproductive technology used to achieve conception. **f**, Gravidity (total number of pregnancies, including the current one) and parity (total number of deliveries, including the current one). **g**, Sampled and regional ambient environmental exposure levels across five major Chinese regions: East China (Zhejiang, Shanghai, Anhui, Jiangsu, Jiangxi, Fujian), West China (Xinjiang Uygur Autonomous Region, Yunnan, Shaanxi), South China (Guangdong, Hainan), Central China (Henan, Hubei, Hunan) and Northeast China (Liaoning). The colored polygons in the radar charts represent population-weighted exposure levels among cohort participants, whereas the violet polygons denote regional environmental averages. Outer donut charts indicate the regional case counts and prevalence of prenatal anxiety and depressive symptoms. Basemap adapted from ref. 35 under a Creative Commons license CC BY 4.0.

oxidative stress, neurotransmitter dysregulation and neurodegenerative processes[21–23]; that extreme heat may elevate mental distress through interacting psycho-physiological pathways involving stress-response activation, thermoregulatory burden, sleep disruption and behavioral disturbance[24,25]; and that green space may confer protection mainly through psychosocial and social–behavioral pathways[19]. Pathological mechanisms underlying mental disorders remain incompletely elucidated; hence, identifying modifiable risk determinants for timely intervention has become one of the most widely employed approaches in observational epidemiology. Considering pregnant mothers as a particularly vulnerable population more susceptible to adverse environmental exposures, these preliminary findings underscore the practical value of integrated, multi-dimensional optimization of residential environments to safeguard maternal health.

However, at the cohort-study level, existing mental health risk assessments have largely focused on general populations, with limited attention given to perinatal women. Besides, the potentials for indirect or interactive effects among various factors, for example, greenery can mitigate urban heat island (UHI) effects[26–28], and complex interactions between endogenous and exogenous determinants can jointly

shape mental health outcomes[29], have sporadically been explored. Furthermore, synergistic or buffering effects across multiple environmental exposures have seldom been systematically reported in observational epidemiological studies[30,31], as conventional epidemiological approaches often examined individual risk factors independently.

Here in this study, leveraging a 13-year, multi-province Chinese maternity cohort[32–35], we investigated how compound city residential environmental exposure jointly shapes maternal prenatal mental health. We first examined how exposure to ambient air pollution, extreme heat and urban greenness were associated with PMDs, with particular attention to extended preconceptional exposure windows and potential interactions between exposures. We then assessed the effect modification by maternal health conditions and behavioral attributes to identify particularly vulnerable subpopulations; and subsequently, we estimated how many PMDs should be attributable to non-optimal environmental exposure. Ultimately, we quantitatively evaluated the composite maternal mental health liveability of all Chinese prefecture-level cities and identified the city-specific predominant adverse environmental exposure to inform locally tailored urban planning and public health strategies. Our findings underscore the salience of prenatal mental health as a pressing public health issue and an essential priority within the broader framework of women's health, marking an appreciable step forward in fortifying women's rights and well-being.

Results

A total of 216,442 pregnant women were recruited for epidemiological investigation across 15 provinces in China during 2013–2025 (Fig. 1a,b), offering a satisfactory level of nationwide representativeness. Through standardized questionnaire-based mental distress pre-screening and subsequent structured diagnostic adjudication across multiple routine prenatal visits, we identified 68,911 cases (31.8%) of PMDs, comprising 34,937 (16.1%) cases of prenatal anxiety symptoms and 33,974 (15.7%) cases of prenatal depressive symptoms. Given that anxiety and depression are not mutually exclusive clinical phenotypes, overlap cases are recorded as 'depressive state with accompanying anxiety features' and classified under depression, as depression carries higher diagnostic priority. In this context, PMDs were selected as the preferred health outcome for risk reporting because they more comprehensively capture the full anxiety–depression symptom spectrum while avoiding fragmented interpretation of co-occurring presentations.

Detailed cohort profiles are illustrated in Fig. 1c–g and Extended Data Table 1. PMD prevalence varied substantially across provinces (Extended Data Table 2). Pregnant women with advantaged sociodemographic profiles tend to reside in neighborhoods with higher-quality green spaces and closer proximity to urban functional areas (Supplementary Tables 1–3), suggesting that not only socioeconomic status probably influences residential location choices but also that places built for more affluent populations may themselves be systematically endowed with superior access to high-quality built-environment resources, including greenness, from the outset.

Phenome-wide risk determinants of maternal mental distress

The determinants of abnormal maternal mental health during pregnancy span socioeconomic and behavioral characteristics, maternal health conditions and environmental exposures (Table 1). Both very young (<20 years) and advanced maternal age (>35 years) were observed to be associated with elevated psychological burden. Our interviews revealed that younger mothers often experienced anxiety stemming from limited awareness and preparedness for pregnancy, whereas women of later reproductive age reported heightened distress exacerbated by pervasive media narratives around 'advanced maternal age' risks. Higher educational attainment (risk ratio (RR) = 0.828, 95% confidence interval (CI): 0.818–0.838), lighter physically or mentally occupational workloads and higher household income (RR = 0.967, 95% CI: 0.955–0.979) were associated with lower prenatal mental risks.

Behavioral characteristics also exhibit risk associations. Representative examples include first-time pregnancy and unplanned conception, which often indicate inadequate psychological preparedness for gestation. Pet companionship during pregnancy significantly alleviated mental stress (RR = 0.801, 95% CI: 0.767–0.836, p-value = 5.74×10^{-24}). Excessive gestational weight gain, as a marker of abnormal dietary behavior, further increased psychological strain, by 5.8% (95% CI: 4.9–6.7%) per unit increase in BMI, predominantly reflecting anxiety about postpartum body-image recovery.

Physiological risk factors primarily include maternal cardiovascular diseases, hepatobiliary conditions, thyroid dysfunction and anemia. Hypothyroidism can directly induce depressive symptoms, while hyperthyroidism often renders expectant mothers more emotionally reactive. According to qualitative feedback from women suffered from multi-symptomatic intrahepatic cholestasis of pregnancy (ICP), manifestations of jaundice and diffuse pruritus[36] served as significant triggers of psychological distress (RR = 1.196, 95% CI: 1.113–1.286). Intakes of folic acid and other essential pregnancy supplements, given near-universal prescription rates in Chinese hospital-based care, were excluded from our cohort analyses.

Risk effects of suboptimal environmental exposures

We identified risk windows for residential ambient environmental exposures spanning from six months before conception to the end of the second trimester, with RRs estimated for PMDs and two clinical subcategories (Fig. 2a–c and Extended Data Fig. 1). Within the periconceptional year, each 10 μg m^{-3} increase in $PM_{2.5}$ exposure and every additional ten extreme heat days (EHDs, defined as days with a daytime maximum hourly humidity-calibrated heat index exceeding 35 °C) were associated with a 14.0% (RR = 1.140, 95% CI: 1.135–1.146) and 16.1% (RR = 1.161, 95% CI: 1.150–1.171) higher risk of PMDs, respectively. Conversely, green space accessibility (GSA) within a 500-m residential buffer was significantly protective against PMDs (RR = 0.877, 95% CI: 0.873–0.881, per 0.1-enhanced vegetation index (EVI) increase), diminishing with increasing distance and disappearing beyond 2 km (Supplementary Fig. 1). No significant cross-regional heterogeneity in the risk associations was observed (Fig. 2d–f).

Residential environments also affect the frequency of outdoor moderate-intensity physical activity (MPA; Supplementary Table 4). $PM_{2.5}$ pollution and heatwaves reduced MPA levels, whereas each 0.1-EVI increase in 500-m GSA corresponded to a 15.9% (95% CI: 15.6–16.2%) higher proportion of women engaging in outdoor exercise. According to self-reports, pregnant women were more inclined to 'go outside' for light activities when greenness appeared within easy walking distance. Greenness offers cleaner air and calming visual cues, fostering both physical and mental relaxation. Women accompanied with animals, particularly dogs, showed nearly threefold higher MPA owing to the routine necessity of dog walking.

The exposure–response curves for the three environmental features are shown in Fig. 2g–i, all displaying monotonic patterns with estimated threshold levels of 18.9 μg m^{-3} for $PM_{2.5}$, 26.6 days for EHDs and 0.125 EVI for GSA. Residential GSA also acted as a buffering factor attenuating the $PM_{2.5}$–PMD risk association. Contrarily, $PM_{2.5}$ pollution and heatwaves exhibited a marked synergistic effect, whereby heat exposure amplified the $PM_{2.5}$–PMD association (Fig. 3a). Although China has a highly developed meteorological warning system and improved housing infrastructure, which effectively prevents direct high-temperature exposure for pregnant women, participants nonetheless articulated indirect stressors associated with extreme heat contributing to mood disturbance during pregnancy: intense sunlight and visual glare, pervasive cicada chirping, restrictions on outdoor mobility and reduced efficiency of urban services such as food delivery or logistics. Moreover, proximity to fixed urban functional and service facilities can also influence maternal mental health (Supplementary Table 5). From the perspective of regional environmental justice, while

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Table 1 | Phenome-wide socio-behavioral and medical risk determinants of PMDs

Risk determinants	Anxiety symptoms		Depressive symptoms		PMDs	
Socioeconomic features	**RR (95% CI)**	***p*-value**	**RR (95% CI)**	***p*-value**	**RR (95% CI)**	***p*-value**
Age at pregnancy (per band)	0.963 (0.946–0.981)***	<0.001	0.956 (0.938–0.975)***	<0.001	0.948 (0.934–0.962)***	<0.001
25–29	Ref.		Ref.		Ref.	
<25	1.575 (1.482–1.673)***	<0.001	3.335 (3.145–3.536)***	<0.001	2.947 (2.803–3.098)***	<0.001
30–34	1.149 (1.115–1.184)***	<0.001	1.446 (1.399–1.494)***	<0.001	1.357 (1.324–1.391)***	<0.001
≥35	1.250 (1.205–1.297)***	<0.001	2.163 (2.086–2.242)***	<0.001	1.922 (1.866–1.979)***	<0.001
Education attainment (per band)	0.916 (0.902–0.930)***	<0.001	0.820 (0.807–0.832)***	<0.001	0.828 (0.818–0.838)***	<0.001
Urban residence (urban/rural)	1.020 (0.981–1.060)	0.32	1.039 (0.997–1.082)	0.07	1.037 (1.005–1.070)*	0.02
Occupation						
Office workers (including white-collar employees)	Ref.		Ref.		Ref.	
Sales, business and technical professionals	1.149 (1.080–1.223)***	<0.001	1.542 (1.454–1.636)***	<0.001	1.478 (1.406–1.554)***	<0.001
Manual laborers (including blue-collar workers)	1.034 (1.006–1.062)*	0.02	1.055 (1.027–1.084)***	<0.001	1.051 (1.029–1.075)***	<0.001
Healthcare professionals	1.229 (0.976–1.549)	0.08	1.816 (1.463–2.254)***	<0.001	1.706 (1.416–2.055)***	<0.001
Education professionals	1.137 (1.044–1.239)**	0.003	1.329 (1.220–1.447)***	<0.001	1.300 (1.213–1.393)***	<0.001
Students	1.354 (0.960–1.911)	0.08	2.478 (1.807–3.399)***	<0.001	2.366 (1.780–3.146)***	<0.001
Self-employment or unemployment	1.093 (1.049–1.139)***	<0.001	1.291 (1.243–1.340)***	<0.001	1.256 (1.214–1.299)***	<0.001
Medical insurance (URRBMI/UEBMI)	1.008 (1.004–1.012)***	<0.001	1.022 (1.018–1.026)***	<0.001	1.019 (1.016–1.022)***	<0.001
Household income level (affluent/constrained)	0.979 (0.964–0.994)**	0.007	0.969 (0.954–0.984)***	<0.001	0.967 (0.955–0.979)***	<0.001
Behavioral features						
Gynecological age (per 5-year increase)	0.955 (0.943–0.968)***	<0.001	0.853 (0.847–0.858)***	<0.001	0.891 (0.887–0.896)***	<0.001
Smoking or second-hand smoke (yes/no)	1.314 (1.004–1.722)*	0.05	1.490 (1.144–1.942)**	0.003	1.575 (1.256–1.975)***	<0.001
Solid fuel use (yes/no)	1.026 (0.970–1.085)	0.37	1.066 (1.008–1.128)*	0.03	1.058 (1.011–1.107)*	0.01
Gestational BMI change (per 1-unit increase)	1.015 (1.005–1.025)**	0.003	1.073 (1.062–1.085)***	<0.001	1.058 (1.049–1.067)***	<0.001
Assisted reproductive technology (yes/no)	1.497 (1.427–1.570)***	<0.001	2.699 (2.581–2.823)***	<0.001	2.497 (2.402–2.595)***	<0.001
Planned conception (yes/no)	0.648 (0.630–0.666)***	<0.001	0.365 (0.356–0.375)***	<0.001	0.392 (0.383–0.401)***	<0.001
3–6 months of preconception preparation (yes/no)	0.645 (0.627–0.663)***	<0.001	0.361 (0.351–0.371)***	<0.001	0.388 (0.380–0.397)***	<0.001
>6-month advanced preconception planning (yes/no)	0.662 (0.634–0.690)***	<0.001	0.387 (0.370–0.404)***	<0.001	0.410 (0.396–0.424)***	<0.001
Pet companionship (yes/no)	0.835 (0.751–0.929)***	<0.001	0.742 (0.697–0.790)***	<0.001	0.801 (0.767–0.836)***	<0.001
Dog companionship (yes/no)	0.839 (0.732–0.961)*	0.01	0.797 (0.735–0.864)***	<0.001	0.826 (0.783–0.872)***	<0.001
Cat companionship (yes/no)	0.889 (0.802–0.985)*	0.02	0.783 (0.723–0.848)***	<0.001	0.850 (0.805–0.898)***	<0.001
Moderate-intensity physical activity (MPA, yes/no)	0.857 (0.842–0.871)***	<0.001	0.712 (0.701–0.724)***	<0.001	0.749 (0.739–0.759)***	<0.001
<150 min per week	Ref.		Ref.		Ref.	
150–300 min per week	0.865 (0.851–0.880)***	<0.001	0.718 (0.706–0.730)***	<0.001	0.757 (0.746–0.767)***	<0.001
>300 min per week	0.772 (0.755–0.790)***	<0.001	0.665 (0.650–0.680)***	<0.001	0.672 (0.660–0.684)***	<0.001
Multipara (yes/no)	0.978 (0.954–1.002)	0.08	0.932 (0.909–0.956)***	<0.001	0.943 (0.924–0.962)***	<0.001
Maternal diseases and medical history						
Maternal cardiovascular diseases (CVDs)	1.013 (0.963–1.066)	0.62	1.253 (1.195–1.315)***	<0.001	1.174 (1.128–1.223)***	<0.001
Maternal primary CVDs	1.143 (1.032–1.266)*	0.01	1.062 (0.955–1.181)	0.27	1.138 (1.045–1.238)**	0.003
Maternal primary cardiac insufficiency	1.231 (0.911–1.663)	0.18	1.539 (1.198–1.977)***	<0.001	1.221 (0.978–1.526)	0.08
Intrahepatic cholestasis of pregnancy (ICP)	1.046 (1.002–1.092)*	0.04	1.176 (1.127–1.227)***	<0.001	1.143 (1.104–1.183)***	<0.001
Single-symptomatic ICP (ICP-S)	1.051 (1.003–1.101)*	0.04	1.144 (1.092–1.199)***	<0.001	1.126 (1.084–1.169)***	<0.001
Multi-symptomatic ICP (ICP-M)	1.052 (0.962–1.150)	0.27	1.246 (1.142–1.360)***	<0.001	1.196 (1.113–1.286)***	<0.001
Thyroid disorders	1.081 (1.016–1.150)*	0.01	1.071 (1.006–1.140)*	0.03	1.099 (1.045–1.156)***	<0.001
Hyperthyroidism	1.127 (1.046–1.214)**	0.002	1.053 (0.859–1.291)	0.62	1.081 (0.924–1.266)	0.33
Hypothyroidism	1.173 (0.973–1.415)	0.09	1.184 (1.100–1.275)***	<0.001	1.210 (1.139–1.286)***	<0.001
Hashimoto's thyroiditis	1.057 (0.932–1.199)	0.39	1.218 (1.066–1.392)**	0.004	1.172 (1.058–1.298)**	0.002
Maternal anemia	1.111 (1.077–1.146)***	<0.001	1.351 (1.310–1.393)***	<0.001	1.302 (1.270–1.335)***	<0.001
Maternal nutritional anemia	1.118 (1.083–1.153)***	<0.001	1.368 (1.326–1.411)***	<0.001	1.318 (1.285–1.352)***	<0.001

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Table 1 (continued) | Phenome-wide socio-behavioral and medical risk determinants of PMDs

Risk determinants	Anxiety symptoms		Depressive symptoms		PMDs	
Socioeconomic features	**RR (95% CI)**	***p*-value**	**RR (95% CI)**	***p*-value**	**RR (95% CI)**	***p*-value**
COVID-19 infection history	1.108 (1.076–1.141)***	<0.001	1.119 (1.092–1.147)***	<0.001	1.099 (1.066–1.133)***	<0.001
History of hepatobiliary diseases	1.258 (1.040–1.522)*	0.02	1.519 (1.266–1.822)***	<0.001	1.565 (1.336–1.834)***	<0.001
History of thyroid surgery	1.006 (0.843–1.201)	0.95	1.255 (1.062–1.483)**	0.008	1.168 (1.014–1.345)*	0.03
Multiple pregnancy	1.169 (1.060–1.289)**	0.002	1.242 (1.130–1.364)***	<0.001	1.320 (1.216–1.433)***	<0.001
Placental implantation disorders	1.058 (0.841–1.332)	0.63	1.096 (1.048–1.146)***	<0.001	1.048 (1.010–1.087)*	0.01
Placenta previa	1.215 (1.028–1.436)*	0.02	1.269 (1.174–1.372)***	<0.001	1.173 (1.098–1.254)***	<0.001
Placenta accreta	1.006 (0.843–1.201)	0.95	1.455 (1.277–1.657)***	<0.001	1.251 (1.116–1.402)***	<0.001

Maternal PMDs (68,911 cases, 31.8% prevalence) under investigation include two subcategories as prenatal anxiety symptoms (34,937 cases, 16.1% prevalence) and depressive symptoms (33,974 cases, 15.7% prevalence). Studied risk determinants cover socioeconomic characteristics, behavioral features and medical diagnoses, with finer subcategories presented in hierarchical indentation. Risk associations, quantified as RRs with 95% CIs, were estimated with multivariable Poisson regression models with robust variance. Two-sided statistical significance levels were annotated as *p-value < 0.05, **p-value < 0.01 and ***p-value < 0.001. All women with a history of major depressive disorder (MDD) exhibited prenatal mental distress symptoms during pregnancy, and therefore history of MDD was treated as a deterministic factor, thus not included in RR assessment. BMI, body mass index; URRBMI, urban–rural resident basic medical insurance; UEBMI, urban employee basic medical insurance. 'Ref.' denotes the reference subgroup/category for comparison in statistical analyses.

suburban and rural residents exhibit certain advantages in natural environmental conditions (for example, lower $PM_{2.5}$ exposure and greater surrounding greenness), they are markedly disadvantaged in accessibility to urban functional facilities and municipal services compared with their urban counterparts (Supplementary Tables 6–8).

Moderation by COVID-19 history and pet companionship

Beyond the additive and counteractive interactions among environmental exposures, we further observed pervasive effect modifications (EMs) on established risk associations towards PMDs by maternal COVID-19 infection history and pet companionship. COVID-19 infection was independently associated with an elevated risk of PMDs (RR = 1.099, 95% CI: 1.066–1.133) and further exacerbated the mental vulnerabilities by 10–20% across three separate high-risk subgroups, compared with their respective reference groups: women (1) who conceived via assisted reproductive technologies (ART), (2) with pre-existing cardiovascular diseases and (3) with pre-pregnancy hematological disorders (Fig. 3b).

Post-acute sequelae of COVID-19 (also referred to as 'long COVID') including persistent fatigue, breathlessness and sleep disturbances emerged as another direct stressor. Moreover, the widespread social speculation about potential teratogenic effects of COVID-19 infection further fueled the psychological burden on the pregnant. In contrast, pet companionship attenuated the relative risks of mental distress induced from placental implantation disorders (for example, placenta previa and placenta accreta spectrum disorders), primary cardiovascular conditions and gestational hypertension by ~10% (RR_i = 0.891–0.924) and amplifying the preventive effects of adequate preconception planning (RR_i = 0.953, 95% CI: 0.924–0.983). With respect to environmental exposures, COVID-19 infection history amplified the mental stress conferred by $PM_{2.5}$ and heatwaves, while simultaneously weakening the relieving effects of GSA. Conversely, pet companionship as another mental soothing behavioral feature manifested the opposite pattern (Fig. 3c,d).

In addition to COVID-19 infection history, experiencing pandemic-related lockdown during pregnancy emerged as another important risk factor for mental distress (RR = 1.138, 95% CI: 1.103–1.175). More specifically, pregnant women experiencing prolonged lockdown (that is, ≥5 months) showed a higher risk of PMDs (RR = 1.196, 95% CI: 1.174–1.218) compared with those exposed to shorter lockdown periods (RR = 1.099, 95% CI: 1.075–1.124). What is worse, experiencing both lockdown and infection revealed a significant combined aggravative effect (RR_i = 1.249, 95% CI: 1.187–1.315), primarily on the depressive symptoms (Supplementary Tables 9–11).

Mental health benefits of greenery through multiple pathways

Synthesizing literature and our findings, we propose that urban greening could substantially reduce PMD burdens through four complementary pathways (Fig. 4), with attributable fractions estimated as follows: (1) mediated via reduction in heatwave frequency[37,38] (26.6%); (2) mediated via increased maternal physical activity during pregnancy (12.9%); (3) moderation through attenuation of the $PM_{2.5}$–PMDs risk association (32.4%) and (4) direct protective effects of green space on mental health together with unspecified residual pathways (28.1%).

Chinese cities exhibit a wide variety of composite residential environmental exposure profiles (Extended Data Fig. 2). Supported by the identified multi-pathway pattern, we further derived a theoretical counterfactual scenario defined as the 'optimal GSA level', in which expanded greenness coverage would offset the mental health burdens associated with suboptimal environmental exposures (Extended Data Fig. 3). We found that prioritizing urban greening represents a feasible and scalable intervention, with ~25% elevating in nationwide population-weighted GSA (that is, from 0.285 to 0.355), capable of buffering the adverse psychological impacts of environmental stressors. The substantial regional heterogeneity in environmental exposure profiles (Fig. 5a) further highlights the need for context-specific and locally adaptive urban planning strategies, as the theoretically optimal greenness increase could reach ~40% in Central and South China.

From a policy perspective, however, optimization of the urban environment should not be limited to greening expansion alone but instead should involve coordinated improvement of air quality, thermal conditions and greenness. This implies that environmental interventions inevitably require differentiated priorities across cities. Compared to a scenario where heatwaves are mitigated at a certain proportion, the corresponding optimal GSA levels demanded were generally lower than that required in another scenario aiming for a comparable proportional $PM_{2.5}$ reduction (Supplementary Fig. 2). Hence in the context of climate change, the ever-increasing frequency of heatwaves appears to have overtaken $PM_{2.5}$ pollution as the principal environmental threat to be addressed in the next phase of liveable and health-supportive city planning—an inflection point greatly attributable to the substantial achievements through China's nationwide 'Blue Sky Protection Campaign' over the past decade.

Nationwide PMDs attributable to environmental exposures

At the individual level, our epidemiological analyses have identified a broad range of risk determinants associated with the occurrence of PMDs. By extension, at the population level, the corresponding socioeconomic and environmental characteristics should also be reflected in area-level variability of PMD prevalences and can therefore serve as credible predictors for ecological estimation. Supported by 707 published studies reporting localized prevalence estimates (Extended Data Fig. 4), we first predicted province-specific PMD prevalences in recent-year China (represented by 2021). We then applied the exposure–response

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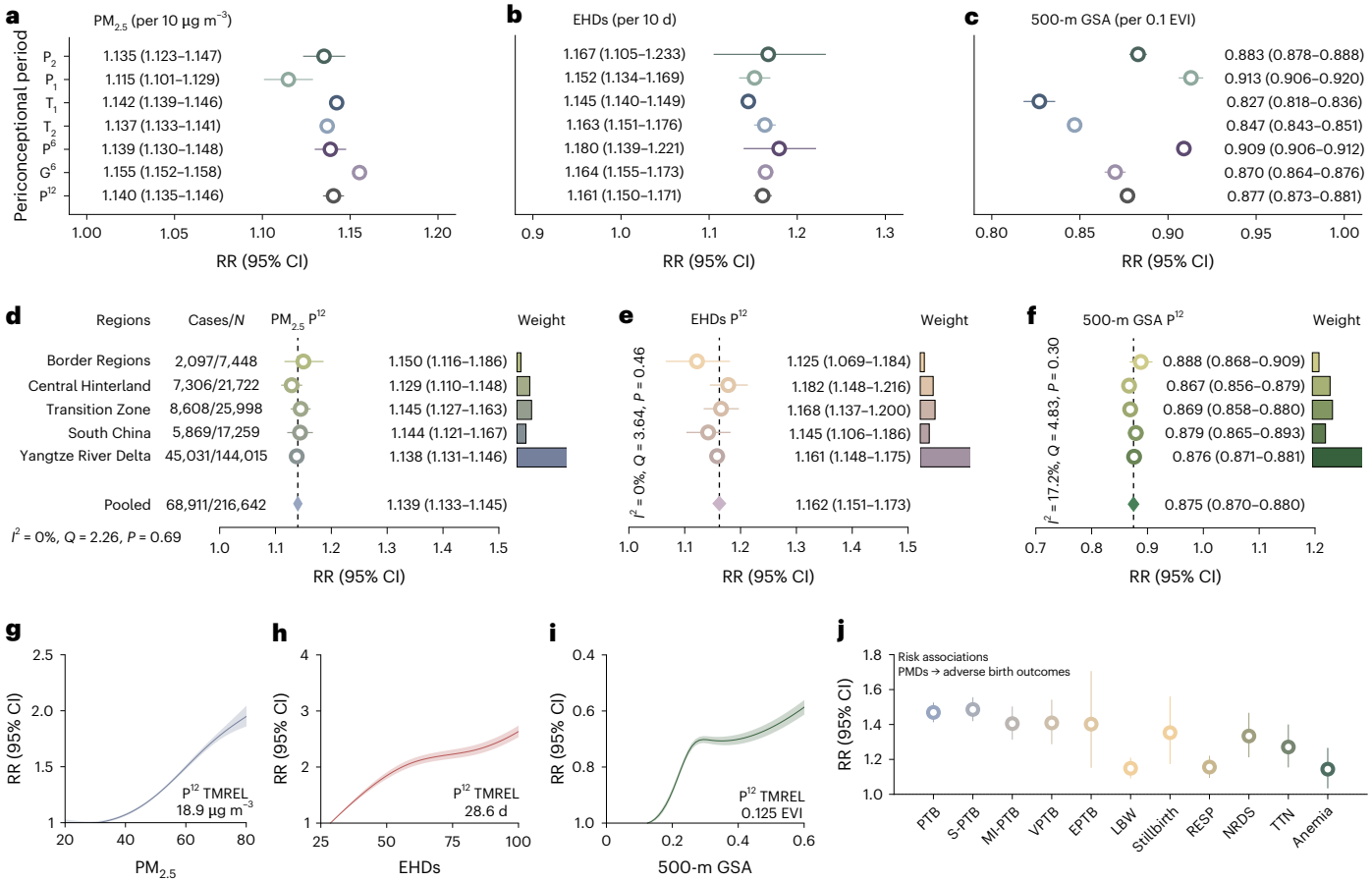


Fig. 2 | Risk associations between ambient environmental exposures, PMDs and adverse birth outcomes. a–c, Associations between prenatal environmental exposures and PMD risk for ambient $PM_{2.5}$ (**a**), EHDs (**b**) and GSA (**c**) within a 500-m residential buffer zone, quantified in RRs. Seven critical periconceptional exposure windows were identified, denoted by subscripted period labels and superscripted duration in months: P_1 (preconception 0–3 months), P_2 (preconception 3–6 months), T_1 (first trimester), T_2 (second trimester), P^6 (preconception 6 months), G^6 (gestational 6 months) and P^{12} (entire periconceptional 12 months). **d–f**, Ambient exposures were quantified by P^{12} averages for $PM_{2.5}$ (**d**), EHDs (**e**) and 500-m GSA (**f**) clustered for five regions: Border Regions (Xinjiang Uygur Autonomous Region, Yunnan and Liaoning), Central Hinterland (Shaanxi, Henan, Hubei and Hunan), Transition Zone (Anhui and Jiangxi), South China (Fujian, Guangdong and Hainan) and Yangtze River Delta (Jiangsu, Zhejiang and Shanghai). Multi-region pooled associations and inter-regional heterogeneity tests based on random-effects meta-analyses by DerSimonian–Laird estimator. **g–i**, Exposure–response curves based on P^{12} averages for $PM_{2.5}$ (**g**), EHDs (**h**), and 500-m GSA (**i**), with estimates of the theoretical minimum risk exposure level (TMREL) indicated. Solid curves display central mean estimates of RR, with shaded bands representing the estimated 95% CIs. **j**, Risk associations between PMDs and a range of adverse birth outcomes (ABOs). PTB, preterm birth; S-PTB, spontaneous preterm birth; MI-PTB, medically indicated (iatrogenic) preterm birth; VPTB, very preterm birth; EPTB, extremely preterm birth; LBW, low birth weight; RESP, neonatal respiratory disorders; NRDS, neonatal respiratory distress syndrome; TTN, transient tachypnea of the newborn. In forest plots (**a–f** and **j**), open circles indicate the central mean estimates of RRs, and whiskers indicate the 95% CIs. Analyses comprised 216,442 participants for **a–c** and **g–j**; variations in the number of participants are stated in **d–f**.

relationships developed in our study to estimate an environmentally attributable net PMD burden of 817,100 (95% uncertainty interval (UI): 701,000–929,500). Specifically, 660,400 (95% UI: 526,100–794,400) cases were attributable to $PM_{2.5}$ exposure, 948,400 (95% UI: 823,000–1,084,000) to heatwave exposure, whereas GSA prevented 791,700 (95% UI: 648,100–948,900) cases (Fig. 6a–f). The estimates suggest that environmentally associated burdens (1.61, 95% UI: 1.35–1.88 million) accounted for nearly half of the all-cause PMDs. Notably, the synergistic effect of concurrent exposure to $PM_{2.5}$ pollution and extreme heat contributes 376,900 (95% UI: 308,100–448,400) cases in total.

By province (Fig. 6g), Guangdong bears the highest environmentally attributable burden as 146,600 (95% UI: 128,800–164,400) cases, predominated by heatwave exposure. The subtropical monsoon climate in Guangdong supports dense vegetative cover, which partially offsets the mental impact of $PM_{2.5}$ pollution. Henan ranks second (100,500, 95% UI: 95,400–104,000 cases), primarily due to persistently high $PM_{2.5}$ pollution and its status as one of the provinces with nationally highest fertility rates. In provinces such as Ningxia and Shanxi with 'high $PM_{2.5}$ pollution, moderate summer temperature and low greenness coverage', the protective effect of GSA roughly neutralized the heat-related risks. In contrast, provinces such as Yunnan with 'low air pollution, mild temperature and high greenness' showed net protective effects as 12,300 (95% UI: 10,700–14,000) cases averted, thus representing ideal geographic settings for gestation under current climatic conditions. The province-level environmental exposure-attributable morbidity rate of PMDs, adjusting for the fertility population-scale effect[39], has also been provided (Supplementary Table 12) for reference.

Micro-scale urban liveability equity

Further adjustment for the cross-sectional prevalence allows the environmental contribution to be more clearly isolated. We therefore developed a composite risk-scoring framework to integrate combined environmental exposures during pregnancy weighted by the estimated strengths of the exposure–response associations, which can be interpreted as a proxy indicator of maternal mental health–related environmental liveability (Fig. 5b). Given the strong epidemiological

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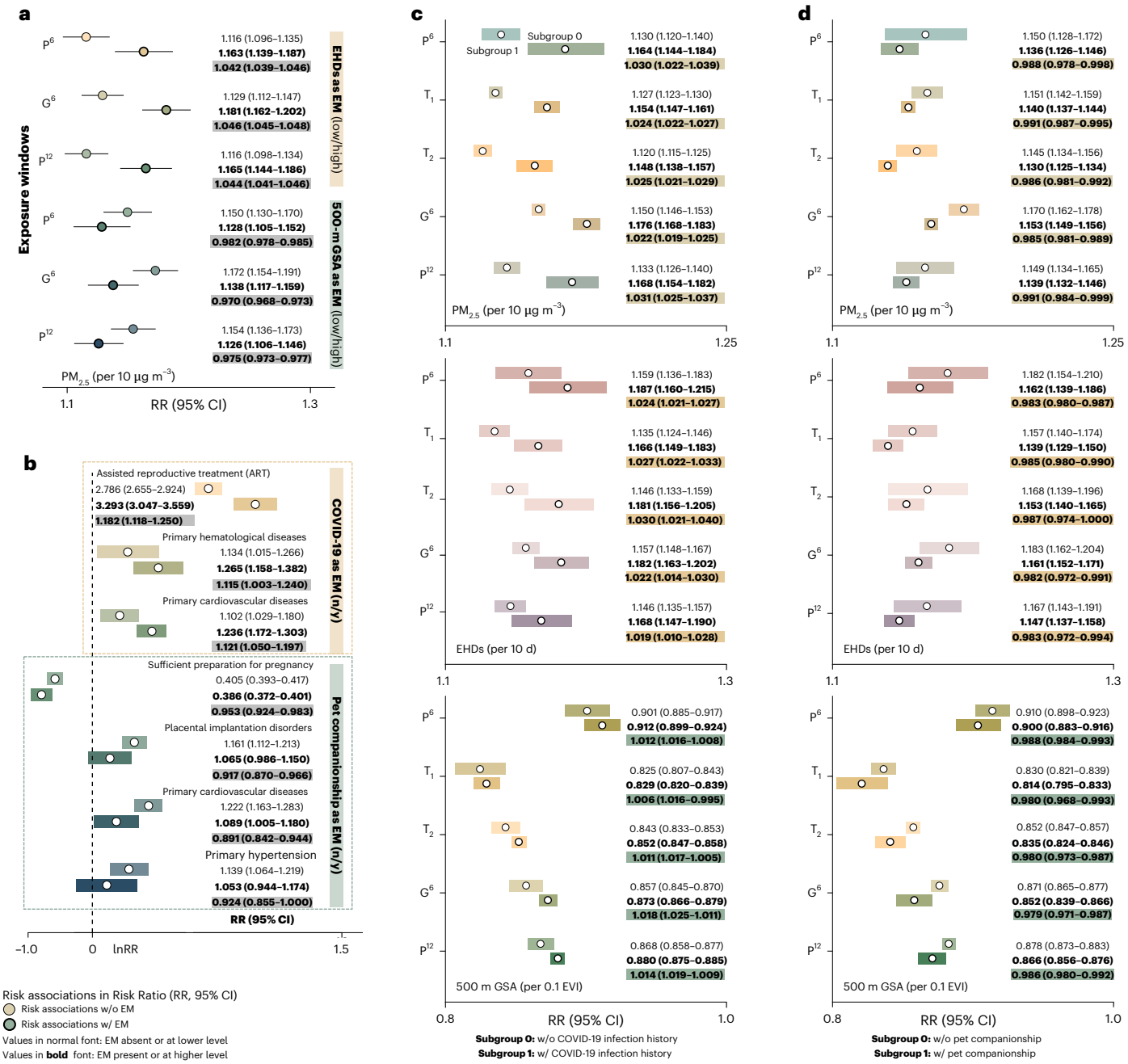


Fig. 3 | Risk interactions between ambient environmental exposures and effect modifications by COVID-19 infection history and pet companionship on risk associations towards PMDs. a, Binary interactions between ambient environmental exposures. Effect modifications (EMs) by exposure to EHDs and GSA within 500-m buffer zone on the association between $PM_{2.5}$ and PMDs were evaluated across three key periconceptional windows (P^6, G^6 and P^{12}). Exposure levels of EHDs and 500-m GSA were dichotomized into high–low categories ($N_H = N_L = 108{,}221$) to illustrate the strength of interaction on a multiplicative scale. **b**, Moderations of the two studied effect modifiers (COVID-19 infection history [$N_0 = 170{,}396$, $N_1 = 46{,}046$] and pet companionship [$N_0 = 139{,}427$, $N_1 = 77{,}015$]) on non-environmental PMD risk determinants. Forest plots show the effect sizes of each risk factor on PMDs, stratified by the presence (y) or absence (n) of each effect modifier. Among all identified risk factors for PMDs in this study, ART, occurrence of primary (pre-existing before conception) cardiovascular diseases, placental implantation disorders and adequate preconception planning showed significant EMs by the two studied effect modifiers (that is, the studied effect modifiers affect the risk associations between the identified risk factors and the occurrence of PMDs). **c**,**d**, Grouped RRs illustrate how ambient environmental exposures during five critical risk windows (P^6, T_1, T_2, G^6 and P^{12}) are modified by COVID-19 infection history (**c**) and pet companionship (**d**), respectively, where 'w/' (with) indicates the presence of the corresponding effect modifier, and 'w/o' (without) indicates its absence. Circles indicate the central mean estimates of RRs (lnRR for **b**); whiskers and the horizontal extent of the colored rectangles indicate the 95% CIs.

links between PMDs and adverse birth outcomes (Fig. 2j), the mental liveability score may also capture broader dimensions of maternal environmental supportiveness to some extent. It therefore has practical city-scale utility for identifying which environmental features should be prioritized for planning intervention (Fig. 6h).

At the intra-city level, an additional question arises as to whether better environmental conditions are accessed by more people. To assess residential equity, we introduced a novel metric, liveability–population alignment (LPA), defined as the synchronicity between grid-level liveability and population density within each city boundary

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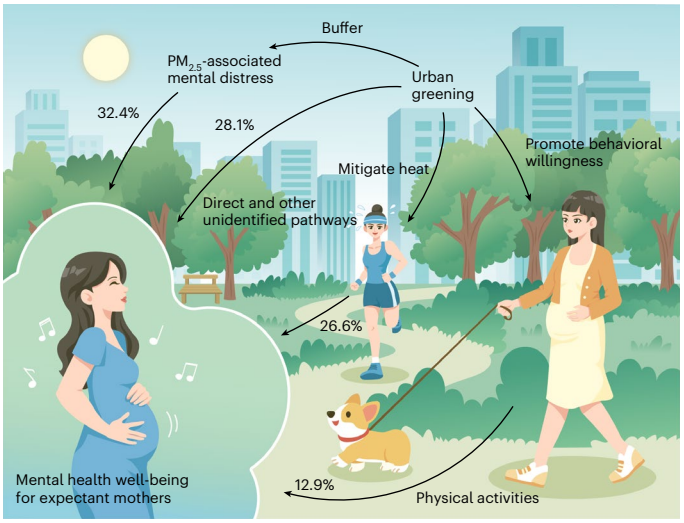


Fig. 4 | Multi-pathway prenatal mental health effects of urban greenery expansion. Multiple pathways through which urban greening mitigates PMDs. Four distinct pathways are quantitatively decomposed stepwise: (I) alleviation of extreme heat, (II) attenuation of $PM_{2.5}$-related mental risks, (III) increased willingness to engage in outdoor physical activity and (IV) the residual contribution interpreted as the direct mental health benefit of urban greening and all the other unexplained mechanisms. First, a counterfactual 0.1-EVI increase scenario relative to current greenness levels yields a direct estimate (DE) jointly capturing Pathways II–IV. Second, Pathway I was quantified independently by recalculating EHDs and attributable burdens under greenness-enhanced cooling. Third, Pathway III was estimated by causal mediation analysis for outdoor physical activity, and Pathway II was isolated by comparing interactive and non-interactive GSA–$PM_{2.5}$ models. Last, the remaining unexplained portion of the DE was assigned to Pathway IV as a residual component. The +0.1-EVI counterfactual scenario was designed because it represents a plausible yet policy-relevant improvement in urban greenness, roughly corresponding to a substantial but still realistic increase in vegetative coverage, while remaining within the observed exposure range of the study population.

(Fig. 5c). The national city-level LPA averages 0.16 (interquartile range (IQR): −0.06 to 0.38; 5–95th percentile: −0.40 to 0.63), with the top 10% cities exceeding 0.55. Strikingly, the spatial distribution of LPA suggests that in most cities, residential clustering patterns are mismatched with local liveable environments; alarmingly, 30.4% of Chinese cities even show negative concordance, where residents tend to live away from environmentally preferable zones.

As environmental liveability scoring was constructed at a fine kilometer-level resolution, LPA can also be calculated at sub-prefecture scales to examine intra-city equity in the distribution of environmental resources. Taking seven Chinese megacities with permanent populations exceeding 10 million as examples, we observed substantial district-level heterogeneity in both liveability and LPA (Extended Data Fig. 5). Although liveability maps may assist a small proportion of affluent residents in planning temporary relocation during specific life stages, governmental and other stakeholders should prioritize universal remediation of residential environments from the perspective of long-term population-health benefits, urban equity and resilience. By conceptualizing these foundational urban health metrics, we propose a planning framework that jointly considers liveability scores and LPA, offering analytical support for health-oriented, place-specific urban policy strategies (Supplementary Box 1).

Discussion

This is a large-scale Chinese multi-province longitudinal maternity cohort study that comprehensively investigates maternal mental health, offering robust primary epidemiological evidence specifically for Pacific Asian populations and enriching the exposure–response literature[40]. Our study has several notable strengths. First and foremost, we undertook a rigorous inquiry into the diverse-spectrum risk determinants of PMDs, encompassing both endogenous socioeconomic and behavioral characteristics and exogenous environmental exposures. Our findings emphasize the critical importance of shifting the window for safeguarding maternal mental health to the preconception period, when early preventive interventions could yield greater psychosocial resilience and mitigate downstream adverse outcomes. By integrating qualitative verbal insights from in-depth interviews, we further contextualized the quantitative associations with subjective risk perceptions of participants.

Next, our study places a strong emphasis on effect modification, highlighting that the health impacts of behavioral characteristics and pre-existing maternal conditions have probably been underestimated in previous literature, as interactions among risk factors are rarely explored. The third spotlight lies in our pioneering attempt to systematically quantify multi-route impacts within a unified analytical architecture, building on the extensive empirical foundation established by prior observational epidemiological research. By enabling the comparability of effect magnitudes across different pathways, this approach represents a notable methodological advance in disentangling complex macro-scale environmental health mechanisms. Our findings through quantitative decomposition are consistent with previous literature suggesting that neighborhood greenness can support psychological resilience partly through encouraging outdoor physical activity[41,42]. In the context of increasingly sedentary urban lifestyles, enhancing nature contact is still of practical relevance for promoting public psychosocial well-being[43,44].

The fourth advantage lies in our multi-factorial attribution framework dissociating the relative contributions of multiple exposures while accounting for the synergistic and buffering interactions, thereby improving the accuracy of burden attribution under complex exposure scenarios, which extends conventional research that typically treats each environmental exposure as an independent factor. Importantly, because the fundamental calculations were performed at high spatial grid resolution, the resulting estimates can be aggregated to any selected arbitrary geographic scale (for example, province, state, county or town) to match the intended study scope. This upscaling design enables the identification of region-specific suboptimal environmental determinants that warrant priority management, therefore offering actionable insights for localized environmental health regulation and urban planning.

From the perspective of urban governance and primary public healthcare, we propose several recommendations to strengthen perinatal mental health management strategies. First, given the already high coverage of routine prenatal visits in China, it is particularly important to strengthen the coverage and frequency of prenatal mental health evaluation together with follow-up services, especially in regions with higher reported prevalence. Earlier and more timely identification of mental distress would facilitate coordinated intervention by hospitals, families and wider social support systems, thereby better safeguarding perinatal health. Second, establish pregnancy-friendly rest and recovery spaces in neighborhoods with limited green space coverage. Third, in areas facing severe $PM_{2.5}$ pollution and heatwave exposures, provide health-supportive intervention kits, including air purifiers and cooling subsidies, to reduce environmental stressors and promote maternal well-being.

Limitations of our study delineate clear directions for future research. First, we acknowledge the exploratory nature of our proposed liveability assessment framework, and the current maternal mental liveability score should only be regarded as an initial 'prototype' for a more encompassing and generalizable overall environmental liveability score in future work. As the framework evolves, it will be imperative to incorporate broader vulnerable populations, a wider constellation of adverse health outcomes and additional environmental exposure

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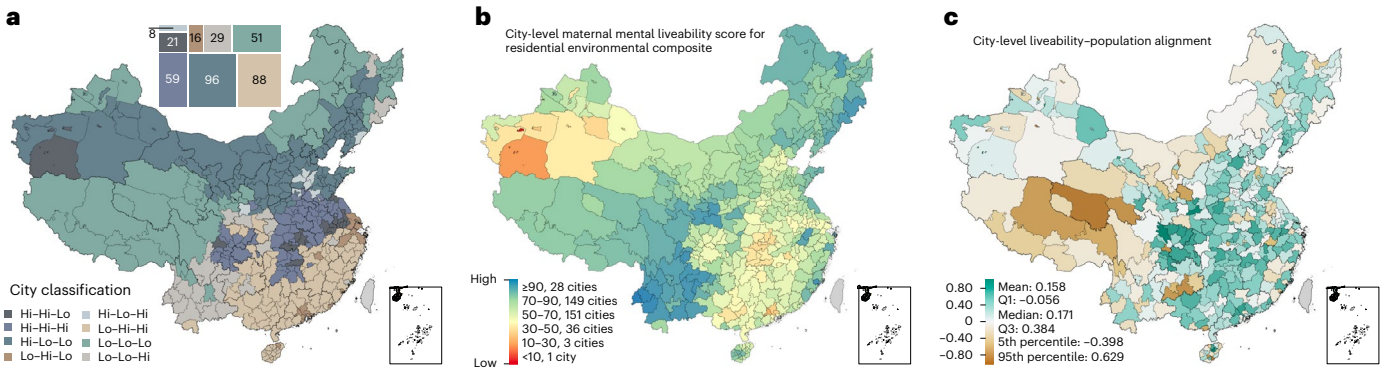


Fig. 5 | City-level composite residential environmental profiles, maternal mental health liveability and spatial equity in population access. a, Three-dimensional environmental archetypes of 368 Chinese prefecture-level cities classified based on $PM_{2.5}$, EHDs and 500-m GSA. Each environmental exposure level by sequence is dichotomized as 'high' (Hi) or 'low' (Lo) based on the nationwide median, thus yielding eight composite categories. For example, 'Hi–Hi–Lo' denotes cities with 'high $PM_{2.5}$ pollution, high EHDs, low GSA'. **b**, City-level scores for maternal mental health liveability defined by residential ambient environmental profiles across China. **c**, City-level liveability–population alignment (LPA). LPA is a novel constructed metric to quantify the spatial concordance between residential environmental quality and population density within each city. Mathematically, LPA is defined as the Pearson linear correlation coefficient between the liveability score and the population density across all grid cells within a city. A higher LPA value indicates a more equitable spatial distribution in which better environmental conditions are shared by a larger proportion of the population. Due to inherent limitations in the correlation-based algorithm, extremely high or low LPA values may be influenced by small numbers of circumscribed grid cells, manifesting in cities with very small land areas in this study. As such, the 5th and 95th percentile range better captures the realistic bounds of LPA values in practice. Basemap adapted from ref. 35 under a Creative Commons license CC BY 4.0.

dimensions (such as visual greenness quality and accessibility to urban functionalities) to improve the comprehensiveness and policy salience of the evaluation system. Second, future works are encouraged to explore unspecified mechanisms beyond the three studied pathways, further elucidating how urban greening can be associated with better mental conditions. Third, fostering cross-provincial and transregional partnerships could strengthen cohort representativeness, identify the origin of spatial heterogeneities and improve the generalizability of the evidence base. Fourth, in Chinese regions with comparatively underdeveloped economies and constrained healthcare capacity, clinical-record digitalization remains limited. Cohort recruitment in these settings often requires handwritten primary clinical archives to be manually digitized before incorporation into electronic databases for analyses, creating intense labor costs. This structural asymmetry has kept participant enrollment in these regions at a nascent stage. Nevertheless, ongoing targeted efforts are progressively redressing this imbalance, with sample accrual from historically underrepresented areas now gaining momentum. Ultimately, a coordinated integration of macro-level epidemiological research, meso-scale real-time tracking of dynamic personal exposure profiles and micro-level pathological enquiries will be essential. As internal exposure assessment technologies advance and become increasingly cost-efficient, high-resolution exposome-based reconstruction and tracing will emerge as a critical next frontier for overcoming current methodological limitations[45]. These complementary perspectives can mutually inform and validate one another, advancing mechanistic insights into the etiology of PMDs and strengthening the evidence base for preventive strategies.

Beyond urban–rural environmental injustice, access to high-quality restorative environments remains unevenly distributed even within urban populations and is often coupled with elevated housing costs and institutional barriers to residence. Consequently, environmentally favorable living conditions are disproportionately enjoyed by higher-income groups, whereas lower-income populations face structural constraints in relocating to healthier environments. From a policy implementation perspective, the lesson of the *Code Red* series launched by *The Hamilton Spectator* instructs that liveability assessments should be adopted to reveal inequality and support targeted public investment, rather than to stigmatize disadvantaged communities and diffuse accountability away from governments and structural systems. We therefore cautiously advocate equity-oriented municipal investment in underserved communities, with suburban areas serving as intermediary nodes for the more balanced provision of healthcare, public transport and green infrastructure across rural and peri-urban interfaces. Suburban development can improve neighboring rural living conditions, accommodate demands from higher-income groups seeking less congested residential alternatives, while also stimulating complementary service sectors such as commerce, healthcare, education and recreation. By leveraging comparatively lower land costs, governments may in turn redirect part of the resulting budgetary savings towards subsidizing temporary life-stage-specific (for example, during the perinatal period) relocation options for lower-income urban residents.

Furthermore, the relevance of pregnancy-friendly environments extends well beyond the provision of rehabilitation- and care-oriented urban services tailored exclusively for pregnant women, as the optimal environmental attributes identified from our present study that benefit maternal mental health represent fundamental building blocks of high-standard healthy cities, with the capacity to simultaneously support a wide collection of vulnerable populations[46–50]. Safeguarding maternal environmental well-being may therefore serve as one of the most sensitive and policy-relevant entry points for advancing healthier and more equitable urban systems. Whereas early-stage screening, supportive care and timely clinical intervention remain paramount for individual-level well-being protection, coordinated action across multiple disciplines, including environmental science, clinical medicine, municipal governance and public health, are essential to translate the evidence-based strategies into scalable, equity-oriented inclusive urban solutions through integrated urban planning, strategic infrastructure investment and adaptive environmental regulatory frameworks. Collectively, by shifting towards proactive, redistributive urban governance, cities can attenuate socioeconomic gradients in environmental health opportunities and foster inclusive, resilient and health-supportive urban living systems.

Methods

Cohort participant recruitment

Building upon previous work[36], ZEBRA maternity cohort enrolled 235,792 pregnant women from January 2017 to June 2025, in addition to 6,237 participants from the 2013–2016 pilot study[51] and further

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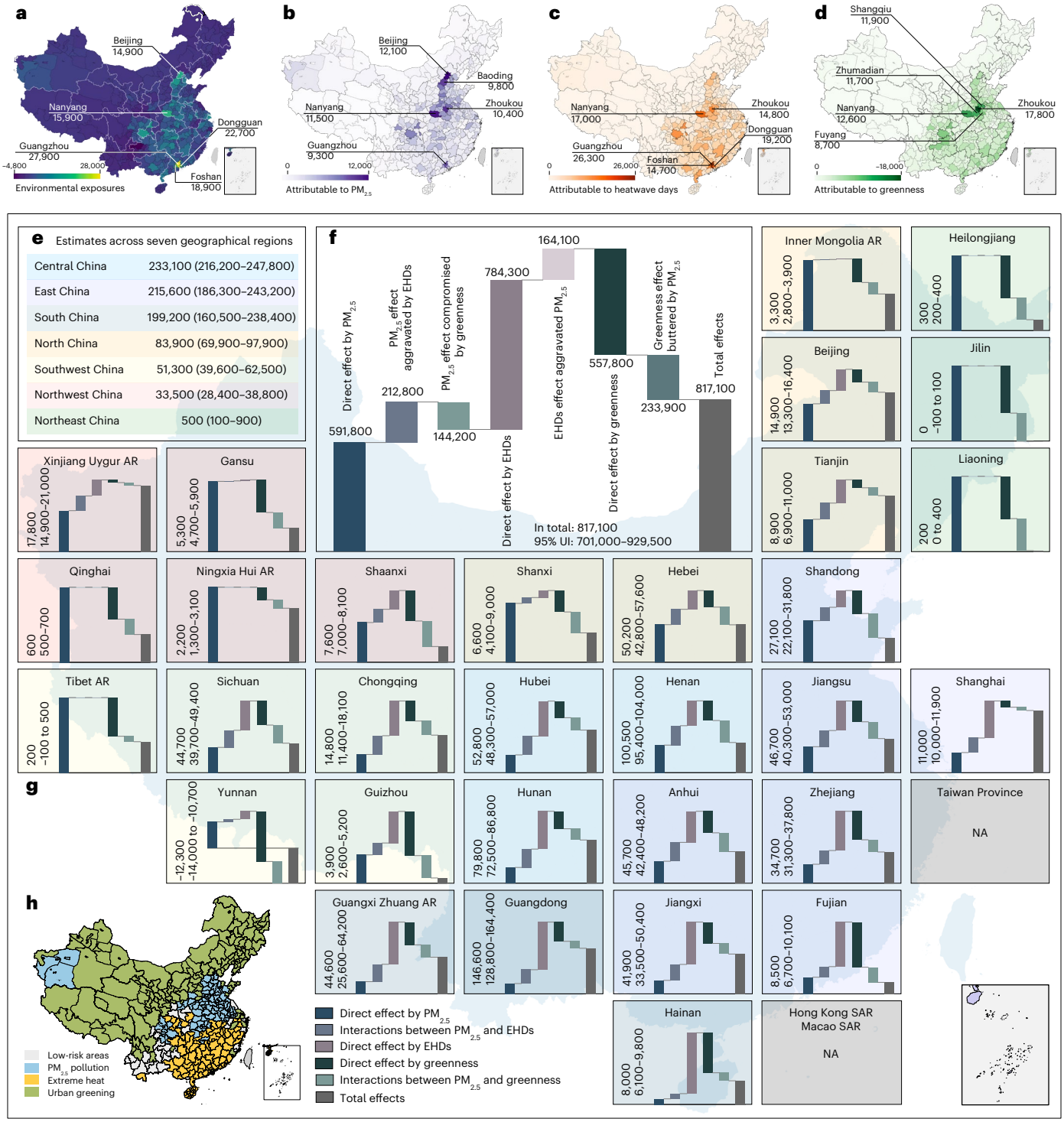


Fig. 6 | City- and provincial-level burden estimates of PMDs attributable to suboptimal environmental exposures. a–**d**, Estimated net PMD burdens associated with overall residential environmental exposures (**a**) and burdens attributable to high $PM_{2.5}$ exposure (**b**), EHDs (**c**) and prevented cases due to residential GSA across Chinese prefecture-level cities (**d**). **e**, Total PMD burdens attributable to the three main environmental exposures aggregated by China's seven major geographical regions, presented in descending order. **f**,**g**, Composition of total attributable PMD burdens for the nationwide (**f**) and provincial (**g**) level. AR, Autonomous Region. **h**, Predominant environmental exposure ('low-risk areas' denote cities where all three exposures fall within low-risk thresholds) that requires priority regulation in each prefecture-level city to safeguard prenatal mental health. Estimated burdens are rounded to the nearest hundred. The Hong Kong Special Administrative Region (SAR), Macao SAR and Taiwan Province were excluded from the analysis due to a lack of consistent underlying data. NA, not applicable due to the unavailability of the underlying dataset for the corresponding region. Basemap adapted from ref. 35 under a Creative Commons license CC BY 4.0.

supplemented by a parallel collaborative sub-cohort comprising 15,226 individuals[32], bringing the full nationwide cohort (Supplementary Fig. 3) to include a total of 257,255 participants. After excluding 19,039 participants without mental health assessments, 20,197 with >20% missing data or unresolvable errors, 64 with gestational age (GA) below 20 or above 44 weeks and 232 participants with maternal age <18 or >45 years, the final analytic cohort consists of 216,442 pregnant women within 15 studied provinces (Fig. 1a).

The study was approved by the Ethics Committee of the Women's Hospital, Zhejiang University School of Medicine (IRB-20220189-R, IRB-20240370-R) and the Institutional Review Board of International Peace Maternity and Child Health Hospital (GKLW2019-11). Written informed consent was obtained from all participants.

Diagnoses of perinatal adverse health outcomes

Initial screening for mental distress and follow-up clinical diagnosis. We employed the Edinburgh Postnatal Depression Scale (EPDS)[52] to assess prenatal mental status across the nationwide cohort. All participants completed EPDS-based mental health pre-assessments during pregnancy, either when the pregnant woman or her accompanying family reported emotional symptoms during routine prenatal visits, when emotional symptoms were observed by outpatient clinicians or within 10 days before the expected date of delivery, thereby also allowing retrospective confirmation of the absence of clinically evident mental distress during pregnancy. As a pre-screening tool, an EPDS score of ≥10 was considered indicative of potential psychological distress, prompting heightened clinical attention and further evaluation through a structured workflow, including: (1) cross-scale screening using Zung's Self-Rating Anxiety Scale and Zung's Self-Rating Depression Scale; (2) joint consultation by a multidisciplinary team involving obstetricians and mental health specialists, and other specialists when clinically required, to distinguish pregnancy-related physiological responses from overlapping psychological symptoms; (3) comprehensive psychological assessment and stratified management; (4) longitudinal follow-up during routine prenatal visits to monitor symptom persistence or progression and (5) postpartum review and final adjudication of symptom-oriented prenatal diagnoses as an 'anxiety state' or 'depressive state', collectively defined as prenatal mental distress symptoms (PMDs). Detailed descriptions of the diagnostic workflow (Supplementary Methods 1), the corresponding flowchart (Supplementary Fig. 4), the distinction between pre-screening results and final clinical diagnoses (Supplementary Methods 2), and sample medical record forms (Supplementary Figs. 5–7) are provided for reference.

Other obstetric and gynecological diagnoses. The maternal clinical conditions considered as potential risk factors included cardiovascular diseases (CVDs) comprising both primary and pregnancy-induced gestational subcategories, intrahepatic cholestasis of pregnancy (ICP) classified into single- and multiple-symptom phenotypes[36], thyroid disorders, anemia and placental implantation disorders. We examined five categories of adverse birth outcomes: stillbirth, preterm birth (PTB), low birth weight (LBW), neonatal respiratory disorders and neonatal anemia. PTB was defined as live birth before 37 completed gestational weeks and further subdivided into spontaneous (S-PTB) and medically indicated subcategories (MI-PTB, also known as iatrogenic PTB). Very (VPTB) and extremely preterm birth (EPTB) were defined as ≤32 weeks and ≤28 weeks of GA, respectively. Neonatal respiratory disorders included both neonatal respiratory distress syndrome (NRDS) and transient tachypnea of the newborn.

Cohort quality control. To ensure the reliability and consistency of clinical diagnoses[33], all clinical staff from participating cities underwent rigorous, standardized training before capturing biopsychosocial profiles and diagnostic information from pregnant participants. Further details on quality assurance procedures for clinical diagnoses are provided in Supplementary Methods 3.

Environmental exposure assessment

Our study evaluated perinatal exposure to three major ambient environmental factors: $PM_{2.5}$, heat and green spaces by geocoding the residential addresses of the cohort participants and projecting onto high-resolution spatiotemporal environmental tracking databases covering 2013–2025. Daily $PM_{2.5}$ concentrations were obtained from the Tracking Air Pollution (TAP) database[53], which integrates multiple data sources including ground-based observations, satellite remote sensing, chemical transport model simulations and land-cover information by means of ensemble machine learning approach to produce highly accurate daily exposure estimates at a 1×1-km^2 resolution.

Heat exposure was assessed by heat index (HI), a humidity-adjusted metric that better reflects the physiological burden of temperature stress[54,55]. Having tested several candidate metrics to characterize extreme temperature events, we originally identified extreme heat days (EHDs), defined as days with maximum hourly HI exceeding 35 °C, as the optimal risk assessment metric, based on statistical robustness and epidemiological relevance. None of the cold-spell-related exposure metrics exhibited statistically significant associations. Temperature and humidity archives were extracted from the European Centre for Medium-Range Weather Forecasts (ECMWF) 5th generation atmospheric reanalysis products (ERA5) at $0.1° \times 0.1°$ spatial resolution[56–58]. Green space accessibility (GSA) was assessed in metric of the enhanced vegetation index (EVI) from the MODIS Vegetation Index Product[59]. EVI was resampled to a 0.5×0.5 km^2 grid and aggregated within 1–3 km buffers surrounding residence of the participants in 16-day temporal intervals.

To improve estimate precision of risk associations, we additionally accounted for individual-level exposure to ambient O_3 and NO_2. These data were retrieved from the TAP[60] and China High-resolution Air Pollution (CHAP) databases[61]. However, as no statistically significant associations were detected for the photochemical pollutants, results for O_3 and NO_2 are not presented as primary findings. For a small proportion of participants enrolled in 2025, beyond the temporal coverage of most environmental tracking databases, residential exposures were approximated using corresponding calendar-day values from the nearest available year, given the strong seasonal cyclicity of these environmental variables.

Statistical analyses and epidemiological inferences

Determination of included covariates for adjustment. Selection of covariates for confounding effect adjustment in the regression models was primarily guided by Directed Acyclic Graphs (DAGs), which were constructed to identify a minimal sufficient adjustment set according to the backdoor criterion, ensuring that all non-causal backdoor paths between environmental exposures and PMDs were blocked while avoiding inappropriate adjustment for mediators or colliders, realized with DAGitty software (https://dagitty.net). In line with the availability of collected information and previous relevant studies (Supplementary Table 13), which have provided identification of known or suspected confounders[19,20,35], a minimum sufficient set of potential confounders for estimating the total effect of environmental exposures on PMDs was determined (Supplementary Fig. 8).

Precision covariates are defined as variables that are associated with the health outcome but not with the exposure. Adjustment for confounding aims to block backdoor paths and obtain unbiased estimates of the exposure effect, whereas inclusion of precision covariates can help explain the outcome, thereby reducing residual variance and improving statistical precision (also known as inferential efficiency) of the estimates. Building upon the cohort data availability, identification of potential precision variables adopted data-driven backward stepwise selection based on Wald tests (inclusion threshold set at p-value < 0.05), regarding PMDs as the health outcome of interest (Supplementary Table 14). All screened candidate variables were subsequently evaluated within the pre-specified DAGs to ensure that mediators or colliders were not inappropriately included in the adjustment set. We finally compiled a collection of candidate covariates to be adjusted by taking the union of two sources[62], and the two-pronged approach ensured both theoretical soundness and data-driven empirical relevance in the final adjustment set.

Poisson regression with robust variance. According to the Bradford Hill causality criteria for temporality, exposure to risk factors should precede the occurrence of health outcomes to support robust association identification. Because the clinical ascertainment of PMDs required persistence of symptoms over at least five consecutive weeks and was often completed during mid-to-late pregnancy, environmental exposures occurring in the third trimester could temporally overlap with symptom emergence and diagnostic ascertainment, thereby complicating risk association interpretation and leading to censoring in observational epidemiological analyses. By the same temporality principle, when pregnancy-related diseases were evaluated as candidate risk determinants, cases with uncertain or relatively late onset were strictly excluded to minimize ambiguity in temporal ordering. Following established research design[35], we focused our exposure windows (EWs) on the periconceptional 12-month interval (P^{12}), spanning from six months before conception (P^6) through to the end of the second trimester (G^6). The rationale for macro-scale environmental health research is supported by converging evidence from both cohort-based observational epidemiology and experiment-based pathological studies, as detailed in Supplementary Methods 4.

We employed Poisson regression with robust variance estimation[63] as the principal analytical approach, given the relatively high prevalence of PMDs in our study population (generally exceeding 10%). In the mathematical form of the regression model, μ represents the absolute risk of event occurrence (that is, prevalence), X denotes the studied risk factors.

$$\log(\mu) = \beta_0 + \sum \beta_i X_i + \sum_{i \neq j} \gamma_{ij} X_i X_j \quad (1)$$

Considering the pronounced interannual periodicity (that is, temporal autocorrelation) in ambient environmental exposures during the periconceptional year, there was no need to control for baseline exposures earlier than the periconceptional year in the Poisson regression models. The relative risk (RR) estimates for each risk factor were subsequently rescaled onto δx, unit change of studied risk factors. Detailed descriptions on the mixed-methods approach for risk interpretation were provided in Supplementary Methods 5.

$$\mathrm{RR} = \exp[\beta \delta x] \quad (2)$$

Interaction and effect modification analyses. We examined three categories of potential interaction effects in this study: (1) interactions between two binary variables (for example, sociodemographic or behavioral attributes, presence of maternal comorbidities); (2) effect modification of continuous ambient environmental exposures by binary stratifying factors and (3) interactions between environmental exposures. To intuitively evaluate and quantify the effect modifications[64], we incorporated multiplicative interaction terms into the Poisson regression model as shown by equation (1), where the coefficient γ represents the log-linear interaction effect between the two variables of interest, X_i and X_j. The magnitude of multiplicative interaction (RR_i) was subsequently derived by:

$$\mathrm{RR}_i = \exp[\gamma_{ij} \delta x_i \delta x_j] \quad (3)$$

Multi-center meta-pooling and meta-regression for heterogeneity source attribution. Given the relatively limited sample sizes in certain provinces, we aggregated the residential locations of cohort participants into five socioeconomically distinct geographical clusters to examine regional heterogeneity in risk associations as main analysis. Supplementary analyses include province-level meta-pooling on crude (Supplementary Figs. 9–11) and adjusted RRs (Supplementary Figs. 12–14). To pool region-specific RRs, we employed the DerSimonian–Laird estimator for random-effects meta-analysis[65]. Cochrane's Q was selected as the primary indicator to assess the between-region heterogeneity, complemented by the I^2 metric to quantify the proportion of total variance attributable to heterogeneity rather than sampling error[66].

We performed meta-regression analyses incorporating a comprehensive set of 60 province-level moderators (Supplementary Table 15), encompassing economic development, demographic structure, socioeconomic characteristics, urban infrastructure, healthcare capacity and population-health awareness to further elucidate the sources of inter-province heterogeneity in RRs (Supplementary Table 16). Provincial archives were extracted from the most recent records of the *China Statistical Yearbook*, *Urban Construction Statistical Yearbook*, *China Environmental Statistical Yearbook* and *China Health Statistical Yearbook*[67].

Exposure–response curved relationship establishment. We employed cubic spline regression models to investigate the nonlinear exposure–response (E–R) relationships between the averaged P^{12} exposure levels of three major environmental factors and the risk of PMDs, where the optimal degrees of freedom were determined according to the Akaike Information Criterion (AIC), and the threshold levels, also known as the theoretical minimum risk exposure levels (TMRELs), were determined by piecewise segmented regression. E–R curves were plotted as supplementary analyses, primarily to (1) assess the monotonicity of the risk associations and to (2) identify potential effect thresholds.

Estimates for province-level prevalence of PMDs

Systematic literature review and data extraction. We searched three databases, the Web of Science, PubMed and China National Knowledge Infrastructure (CNKI, in Chinese language) for peer-reviewed articles reporting regional prevalences of PMDs in mainland China since 2000, searching by terms 'prenatal depression' and 'prenatal anxiety'. Initial searching identified a total of 2,922 studies; and after rigorous multistage screening processes, 707 studies were ultimately retained for data extraction, comprising 336 focused on anxiety symptoms and 371 on depressive symptoms. Two researchers conducted the screening and selection process, assisted by DeepSeek and Qwen LLMs, to identify eligible studies and cross-validate inclusion decisions. In cases of disagreement or uncertainty, final adjudication was made by the principal investigator. Details for the inclusion or exclusion criteria are provided in the Supplementary Methods 6 and Extended Data Fig. 4. Studies providing site-specific prevalences of multiple regions within a province were treated as separate observations. Studies reporting aggregated prevalences spanning multiple years were assigned to the midpoint year of the study period. To avoid robustness compromission by spatiotemporal sparsity, we aggregated all eligible literature subsets published during 2015–2024 to represent the 'recent-year' cross-sectional estimate for downstream modeling.

Province-level prevalence prediction. In this study, we applied a Bayesian multi-level logistic regression model with province-specific random intercepts and random time slopes to estimate annual prevalences of PMDs[68]. We employed the newly developed integrated nested Laplace approximation (INLA) approach for Bayesian inference (https://www.r-inla.org/what-is-inla) in the wide framework of latent Gaussian models (for example, Generalized Linear Mixed Models in our study) owing to high computational efficiency, stability and accuracy[69]. On the basis of prior research identifying critical determinants of PMDs, we selected ten features as predictors in the model: five socioeconomic factors (gross domestic product per capita, urbanization rate, employment rate, higher education attainment rate and number of localized hospitals), three environmental factors (temperature, $PM_{2.5}$ concentrations and greenness coverage) and the incidence rates of two adverse pregnancy outcomes: PTB and LBW. All predictors were clustered onto annually provincial averages,

with sources listed in Supplementary Methods 7. Details for the model can be found in Supplementary Methods 8. The predicted posterior distributions of province-specific prevalences were summarized in mean with 95% Bayesian credible intervals (BCIs) as presented in Supplementary Table 17. We retained 20% of the literature-based records for evaluating the predictive performance (Supplementary Table 18), primarily focusing on three key metrics: coefficient of determination (R^2), root mean square error and coverage rate defined as the proportion of observations falling within the 95% BCIs of the posterior estimates[70].

Suboptimal environmental exposure-attributable burden estimation

Given that the trends shown by E–R curves did not violate the assumption of effect monotonicity and that curve-based fitting, at the current stage, was limited in its ability to quantify cross-exposure interactions, we ultimately based the population-level environment-associated burden estimation of PMDs on the TMRELs derived from the E–R curves, the linear RRs estimated from all participants and across the P^{12} EW through which statistical significances were observed consistently, with multiplicative interaction terms included. The hazardous exposure increment (Δx) was defined from exposure level x by:

$$\Delta x = \max_{\geq 0}(x - \mathrm{TMREL}) \tag{4}$$

To account for interaction effects, we calculated the modified exposure-dependent relative risks (RR_{x_y}) by:

$$\mathrm{RR}_{x_y} = \exp\left[\ln \mathrm{RR}\frac{\Delta x}{\delta x} + \ln \mathrm{RR}_i \frac{\Delta x}{\delta x}\frac{\Delta y}{\delta y}\right] \tag{5}$$

where x_y denotes the risk association of exposure x under the effect modification by exposure y, with unit RRs derived from Poisson regression model. The population attributable fractions (PAF)[71] for each environmental determinant were then estimated by:

$$\mathrm{PAF}_x = \frac{\mathrm{RR}_x - 1}{\mathrm{RR}_x} \tag{6}$$

The exposure-attributable burden of PMDs was quantified by equation (7), incorporating the prevalence (P_{PMDs}), location-specific population (Pop^l) sourced from the LandScan population data platform (https://landscan.ornl.gov) resolved at 1×1 km^2 and updated with city-level calibration[72] and province-specific fertility rates (R_f).

$$\Delta\mathrm{PMDs}_x^l = \mathrm{Pop}^l \times R_\mathrm{f}^l \times P_{\mathrm{PMDs}}^l \times \mathrm{PAF}_x^l \tag{7}$$

Finally, mutually exclusive attributable burdens from multiple environmental risk factors i were summed to yield the total burden associated with non-optimal ambient environmental exposures, followed by location-specific population attributable fraction (PAF^l) adjusting contributions from non-environmental factors.

$$\Delta\mathrm{PMDs}^l = \sum_i \Delta\mathrm{PMDs}_{x_i}^l \tag{8}$$

$$\mathrm{PAF}^l = \frac{\Delta\mathrm{PMDs}^l}{\mathrm{Pop}^l \times R_\mathrm{f}^l \times P_{\mathrm{PMDs}}^l} \tag{9}$$

Provincial fertility rates were approximated by reported birth rates from the China Health Statistical Yearbook[67]. To characterize uncertainty in the burden estimates, we employed a 10,000-realization Monte Carlo bootstrap simulation from which the arithmetic means and 95% UIs were extracted.

Enhanced extensive quantitative analyses

Greenness-mediated burden decomposition and counterfactual optimization. Considering the observed cooling effects of urban greenness[26,27,37,73], we implemented ensemble learning to predict the theoretical ambient temperatures under a range of counterfactual vegetation scenarios (Supplementary Methods 9). Building upon the main attributable burden estimation analysis, we further dissociated the mental health benefits of greenness into multiple complementary pathways through step-by-step decomposition (Supplementary Methods 10). Subsequently, we estimated the optimal GSA level necessary to offset prenatal mental health burdens associated with adverse environmental exposures (Supplementary Methods 11), supported by a simplified greenness–heatwave response relationship derived from parametric curve fitting (Supplementary Fig. 15), which is a function-based extension for Extended Data Fig. 2d. Finally, we fitted quadratic isopleths[74] of optimal GSA levels onto $\mathrm{PM}_{2.5}$ and EHDs under variable exposure scenarios, as visualized in Supplementary Fig. 2, to guide policy prioritization for suboptimal environmental exposure regulation (Supplementary Methods 12).

Other supplementary analyses. To contextualize our main findings, we assessed the association between temporal overlap with China's dynamic COVID-19 control period and PMD risk, including timing and duration of lockdown exposure (Supplementary Methods 13). We further evaluated social inequity in access to high-quality residential environmental resources by comparing green space quality and municipal-resource accessibility across urban–suburban–rural contexts and maternal socioeconomic groups, supported by Essential Urban Land Use Categories Map of China[75] (Supplementary Methods 14).

Quantitative assessment for mental health liveability and equity

To quantify city-specific environmental burdens on maternal mental health, we first calculated the environmentally attributable fraction for each prefecture-level city ($\mathrm{PAF}^{\mathrm{city}}$) by dividing the city-aggregated total burdens attributable to suboptimal ambient environmental exposures by the total number of PMD cases:

$$\mathrm{PAF}^{\mathrm{city}} = \frac{\sum_{\mathrm{city}} \Delta\mathrm{PMDs}^l}{\sum_{\mathrm{city}} \mathrm{Pop}^l \times R_\mathrm{f}^l \times P_{\mathrm{PMDs}}^l} \tag{10}$$

Next, $\mathrm{PAF}^{\mathrm{city}}$ was normalized across all cities to derive a composite 0–100 score scaled index:

$$\mathrm{score}^{\mathrm{city}} = \frac{\mathrm{PAF}_{\max}^{\mathrm{city}} - \mathrm{PAF}^{\mathrm{city}}}{\mathrm{PAF}_{\max}^{\mathrm{city}} - \mathrm{PAF}_{\min}^{\mathrm{city}}} \times 100 \tag{11}$$

The integrated metric effectively characterizes the pure environmental effect, excluding confounding influences from population size, fertility rate and baseline prevalence of PMDs and thus defined as the 'liveability score'. In the preceding sections, both the geospatial mapping of individual-level environmental exposures and the estimation of population-level exposure-attributable health burdens were performed on high-resolution spatial grids as the basic analytical unit. Only in the present section were the grids aggregated to the city level for a more summary-oriented analysis. Hence the liveability score can also be calculated at location-specific grid-scale (s^l):

$$s^l = \frac{\mathrm{PAF}_{\max}^l - \mathrm{PAF}^l}{\mathrm{PAF}_{\max}^l - \mathrm{PAF}_{\min}^l} \times 100 \tag{12}$$

We further constructed an intra-city-scale metric termed liveability–population alignment (LPA) to quantify the spatial equity of high-quality residential environmental features. LPA was defined as the

Pearson correlation between grid-level liveability score and population density within each city boundary:

$$\mathrm{LPA} = \rho\left(s^l, \mathrm{Pop}^l\right) = \frac{\sum_{\mathrm{city}}\left(s^l - \bar{s}\right)\left(\mathrm{Pop}^l - \bar{\mathrm{Pop}}\right)}{\sqrt{\sum_{\mathrm{city}}\left(s^l - \bar{s}\right)^2}\sqrt{\sum_{\mathrm{city}}\left(\mathrm{Pop}^l - \bar{\mathrm{Pop}}\right)^2}} \quad (13)$$

Sensitive analysis

We conducted multiple sensitivity analyses to ensure the robustness of our epidemiological inferences. These included: (1) randomly selecting 50% of the original sample and repeating the statistical analysis five times to test consistency (Supplementary Table 19); (2) adjusting for different combinations of environmental exposures to evaluate the effects of potential confounding structures (Supplementary Table 20) and (3) excluding the collaborative parallel sub-cohort to assess whether the estimated risk associations were altered (Supplementary Table 21).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Individual-level records of ZEBRA maternity cohort participants, including socioeconomic characteristics, residential addresses, medical histories and current pregnancy-related health status, were collected across multiple cities in China. Owing to ethical and legal restrictions and the need to protect participant privacy, these individual-level data cannot directly be available to the public. However, an appropriately anonymized and de-identified version of the dataset, limited to the variables necessary to reproduce and verify the analyses reported in the present study, has been deposited via Github at https://github.com/csuen27/zebra-maternity-cohort. The repository will be actively maintained and updated in response to research-based requests for 12 months following publication. Researchers seeking further access to the ZEBRA maternity cohort should submit a brief research proposal to the principal investigators and corresponding authors, H.Z.S. or H.H., for review. Requests will usually receive a response within four weeks. Following approval by the ZEBRA Collaborative Committee and the Health Commission of Zhejiang Province, restricted access can be granted for research collaboration purposes for reasons of participant confidentiality and data-sharing agreements with the Chinese Medical Association (CMA). For urban residential environmental exposure assessment, daily high-spatial-resolution ambient $PM_{2.5}$ and O_3 concentrations can be obtained from the Tracking Air Pollution (TAP) database (http://tapdata.org.cn)[53]; ambient NO_2 concentrations are provided by the China High-resolution Air Pollution (CHAP) database[61]; ambient temperature and humidity archives can be retrieved from the ECMWF ERA5 reanalysis products (https://www.ecmwf.int/en/forecasts/datasets)[56–58]; the enhanced vegetation index (EVI) can be derived from the MODIS Vegetation Index Product (https://modis.gsfc.nasa.gov/data/dataprod)[59]; the geographical distribution of municipal functional facilities can be supported by the Essential Urban Land Use Categories Map across China (EULUC China 2.0)[75]; and spatially resolved population density can be sourced from the LandScan population data platform (https://landscan.ornl.gov).

Code availability

Analytical codes directly associated with cohort participant information can be made available from the corresponding author(s) upon reasonable request. To support the reproducibility of the core findings, a fully transparent and executable analysis codebase, together with accompanying analytical outputs, has been documented in Jupyter Notebooks and deposited via GitHub at https://github.com/csuen27/zebra-mental-liveability.

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Author contributions

H.Z.S., X.B., Y.W. and H.H. conceptualized the study and oversaw the administration of the maternity cohort. Y.T., H.T., Z.W., Q.X., H. Liu, C.Z., X.L., H. Li, Y.G., Y. Xu, H. Lao, Yuhua Chen, J.Z., R.Y., J. Guo, Y.Y., X.Y. and other members of the ZEBRA and PHAS collaborative research network collected, curated and pre-processed the medical records and associated data. H.C., J. Gao and Yiting Chen conducted the systematic review and data integration. H.Z.S. performed statistical analyses and drafted the paper with substantial support from C.D. J.H., Y. Xie and Y.L. scrutinized and edited the paper. Senior authors G.G.L., L.P.S., B.S., N.W., T.L. and H.H. supervised the study design and provided critical review and editing of the paper.

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Competing interests
The authors declare no competing interests.

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Zhejiang Environmental and Birth Health Research Alliance (ZEBRA) Collaborative Research Network

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Planetary Health Axis System (PHAS) Collaborative Research Network

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Extended Data Table 1 | Descriptive summary of ZEBRA maternity cohort profile characteristics

Characteristics	Overall		Good mental health		Anxiety symptoms		Depressive symptoms	
	N=216,442		N=147,531		N=34,937		N=33,974	
Socioeconomic features								
Age at pregnancy	31.2	4.3	30.9	3.8	31.3	4.5	32.3	5.6
<25	8,672	4.0%	4,598	3.1%	1,757	5.0%	2,317	6.8%
25–29	74,134	34.3%	52,556	35.6%	11,664	33.4%	9,914	29.2%
30–34	87,040	40.2%	64,762	43.9%	12,624	36.1%	9,654	28.4%
≥35	46,596	21.5%	25,615	17.4%	8,892	25.5%	12,089	35.6%
Highest education attainment level								
High school or below	26,837	12.4%	16,175	11.0%	4,960	14.2%	5,702	16.8%
College education	73,262	33.8%	48,496	32.9%	12,439	35.6%	12,327	36.3%
Undergraduate	78,829	36.4%	56,142	38.1%	11,883	34.0%	10,804	31.8%
Graduate or above	37,514	17.3%	26,718	18.1%	5,655	16.2%	5,141	15.1%
Urban residence	173,414	80.1%	118,434	80.3%	27,925	79.9%	27,055	79.6%
Occupation								
Office workers (including white-collar employees)	7,813	3.6%	4,559	3.1%	1,453	4.2%	1,801	5.3%
Sales, business, and technical professionals	102,996	47.6%	71,560	48.5%	16,186	46.3%	15,250	44.9%
Manual laborers (including blue-collar workers)	79,445	36.7%	53,084	36.0%	13,246	37.9%	13,115	38.6%
Healthcare professionals	2,662	1.2%	1,558	1.1%	492	1.4%	612	1.8%
Education professionals	8,485	3.9%	5,516	3.7%	1,461	4.2%	1,508	4.4%
Students	805	0.4%	412	0.3%	163	0.5%	230	0.7%
Others	14,236	6.6%	10,842	7.3%	1,936	5.5%	1,458	4.3%
High household income level	79,223	36.6%	60,573	41.1%	10,182	29.1%	8,468	24.9%
Behavioral features								
Smoking or second-hand smoke (SHS)	3,574	1.7%	2,001	1.4%	746	2.1%	827	2.4%
Household solid fuel use	18,061	8.3%	11,464	7.8%	3,139	9.0%	3,458	10.2%
Gestational BMI change	5.20	1.18	5.16	1.14	5.24	1.13	5.34	1.38
Assisted reproductive technology (ART)	12,963	6.0%	6,956	4.7%	2,618	7.5%	3,389	10.0%
Planned conception	164,951	76.2%	119,767	81.2%	24,317	69.6%	20,867	61.4%
3–6 months of preconception preparation	124,386	57.5%	92,555	62.7%	17,701	50.7%	14,130	41.6%
>6-month advanced preconception planning	40,565	18.7%	27,212	18.4%	6,616	18.9%	6,737	19.8%
Pet companionship	77,015	35.6%	54,970	37.3%	11,835	33.9%	10,210	30.1%
Dog companionship	39,785	18.4%	28,178	19.1%	6,181	17.7%	5,426	16.0%
Cat companionship	38,717	17.9%	27,815	18.9%	5,885	16.8%	5,017	14.8%
Moderate-intensity physical activity (MPA)	166,308	76.8%	119,716	81.1%	25,360	72.6%	21,232	62.5%
<150 min/week	50,134	23.2%	27,815	18.9%	9,577	27.4%	12,742	37.5%
150–300 min/week	104,800	48.4%	74,091	50.2%	16,817	48.1%	13,892	40.9%
>300 min/week	61,508	28.4%	45,625	30.9%	8,543	24.5%	7,340	21.6%
Multipara	91,998	42.5%	62,923	42.7%	14,824	42.4%	14,251	41.9%
Maternal diseases and medical history								
Maternal cardiovascular diseases (CVDs)	12,774	5.9%	7,981	5.4%	2,169	6.2%	2,624	7.7%
Maternal primary CVDs	2,562	1.2%	1,670	1.1%	466	1.3%	426	1.2%
Maternal gestational CVDs	10,212	4.7%	6,311	4.3%	1,703	4.9%	2,198	6.5%
Maternal pregnancy-induced cardiac insufficiency	385	0.2%	234	0.2%	55	0.2%	96	0.3%
Maternal preeclampsia	3,347	1.5%	1,929	1.3%	581	1.7%	837	2.5%
Intrahepatic cholestasis of pregnancy (ICP)	18,307	8.5%	12,018	8.1%	3,063	8.8%	3,226	9.5%
Single-symptomatic ICP (ICP-S)	14,472	6.7%	9,567	6.5%	2,410	6.9%	2,495	7.3%

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Extended Data Table 1 (continued) | Descriptive summary of ZEBRA maternity cohort profile characteristics

Characteristics	Overall		Good mental health		Anxiety symptoms		Depressive symptoms	
	N=216,442		N=147,531		N=34,937		N=33,974	
Multi-symptomatic ICP (ICP-M)	3,835	1.8%	2,451	1.7%	653	1.9%	731	2.2%
Thyroid disorders	8,075	3.7%	5,226	3.5%	1,423	4.1%	1,426	4.2%
Hyperthyroidism	815	0.4%	542	0.4%	150	0.4%	123	0.4%
Hypothyroidism	5,349	2.5%	3,321	2.3%	986	2.8%	1,042	3.1%
Hashimoto's thyroiditis	1,911	0.9%	1,363	0.9%	287	0.8%	261	0.8%
Maternal anemia	37,465	17.3%	23,885	16.2%	6,491	18.6%	7,089	20.9%
Maternal nutritional anemia	36,296	16.8%	23,045	15.6%	6,317	18.1%	6,934	20.4%
COVID-19 infection history	46,046	21.3%	30,070	20.4%	8,068	23.1%	7,908	23.3%
History of hepatobiliary diseases	12,299	5.7%	8,324	5.6%	2,001	5.7%	1,974	5.8%
History of thyroid surgery	976	0.5%	607	0.4%	167	0.5%	202	0.6%
Multiple pregnancy	2,887	1.3%	1,530	1.0%	615	1.8%	742	2.2%
Placental implantation disorders	16,125	7.5%	10,452	7.1%	2,656	7.6%	3,017	8.9%
Placenta previa	4,434	2.0%	2,737	1.9%	741	2.1%	956	2.8%
Placenta accreta	7,156	3.3%	4,579	3.1%	1,200	3.4%	1,377	4.1%
Adverse birth outcomes								
Preterm birth (PTB)	20,455	9.5%	12,271	8.3%	3,616	10.4%	4,568	13.4%
Spontaneous preterm birth (S-PTB)	13,927	6.4%	8,541	5.8%	2,442	7.0%	2,944	8.7%
Medically indicated preterm birth (MI-PTB)	6,528	3.0%	3,730	2.5%	1,174	3.4%	1,624	4.8%
Very preterm birth (VPTB)	3,529	1.6%	2,146	1.5%	608	1.7%	775	2.3%
Extreme preterm birth (EPTB)	748	0.3%	478	0.3%	113	0.3%	157	0.5%
Low birth weight (LBW)	11,775	5.4%	7,487	5.1%	2,065	5.9%	2,223	6.5%
Stillbirths	1,385	0.6%	859	0.6%	236	0.7%	290	0.9%
Neonatal respiratory disorders	9,703	4.5%	6,331	4.3%	1,604	4.6%	1,768	5.2%
Neonatal respiratory distress syndrome (NRDS)	2,819	1.3%	1,733	1.2%	495	1.4%	591	1.7%
Transient tachypnea of the newborn (TTN)	2,797	1.3%	1,779	1.2%	453	1.3%	565	1.7%
Neonatal anaemima	2,771	1.3%	1,805	1.2%	472	1.4%	494	1.5%

Continuous variables (age at pregnancy and gestational BMI change) are summarized as arithmetic means with standard deviations, while categorical variables (indicating presence or absence of a condition/event) are presented as counts with percentages. Statistics are grouped by overall cohort participants, participants with good mental conditions, participants diagnosed with prenatal anxiety symptoms and prenatal depressive symptoms.

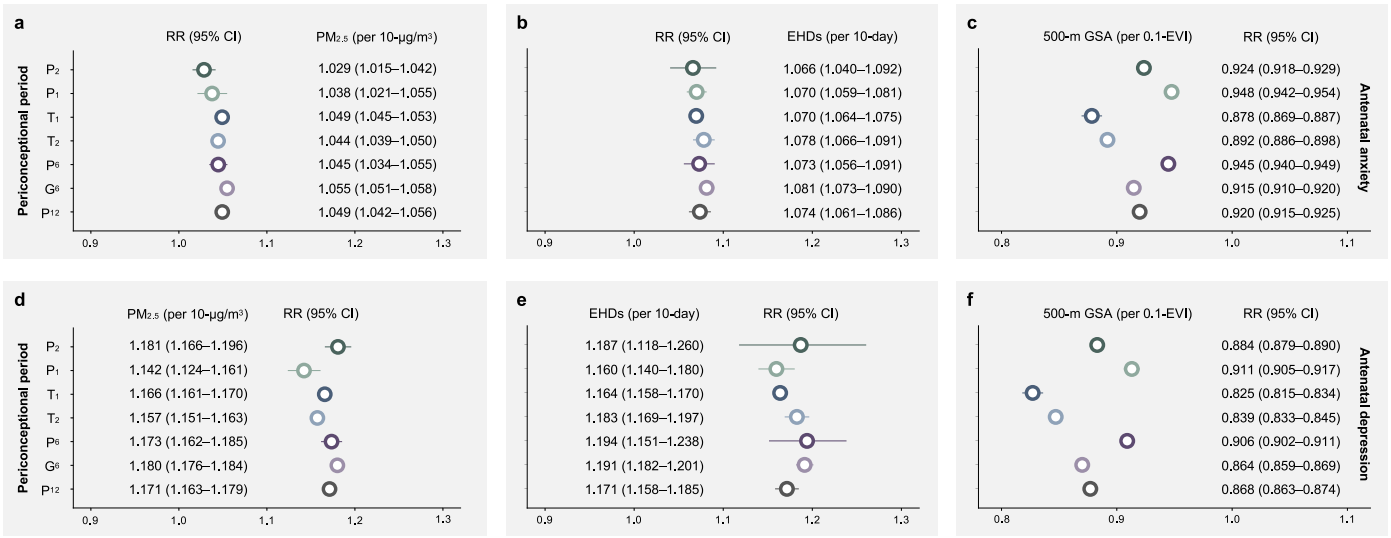
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Extended Data Table 2 | Provincial participant recruitment and prevalence of prenatal mental distress symptoms (PMDs) for ZEBRA maternity cohort

Province	Participants	Anxiety symptoms	Depressive symptoms	PMDs	Prevalence of PMDs
Xinjiang Uygur Autonomous Region	1,217	199	167	366	30.07% (27.56–32.71%)
Shaanxi Province	2,440	379	391	770	31.56% (29.74–33.43%)
Liaoning Province	1,636	244	201	445	27.20% (25.10–29.41%)
Henan Province	5,705	876	940	1,816	31.83% (30.64–33.05%)
Hainan Province	4,228	761	655	1,416	33.49% (32.08–34.93%)
Anhui Province	17,032	3,150	2,581	5,731	33.65% (32.94–34.36%)
Fujian Province	7,029	1,372	1,279	2,651	37.72% (36.59–38.85%)
Guangdong Province	6,002	906	896	1,802	30.02% (28.88–31.20%)
Hubei Province	6,858	1,127	1,135	2,262	32.98% (31.88–34.11%)
Hunan Province	6,719	1,256	1,202	2,458	36.58% (35.44–37.74%)
Jiangsu Province	9,643	1,812	1,691	3,503	36.33% (35.37–37.29%)
Jiangxi Province	8,966	1,447	1,430	2,877	32.09% (31.13–33.06%)
Shanghai	8,736	1,198	1,128	2,326	26.63% (25.71–27.56%)
Yunnan Province	4,595	637	649	1,286	27.99% (26.71–29.30%)
Zhejiang Province	125,636	19,573	19,629	39,202	31.20% (30.95–31.46%)
Overall	**216,442**	**34,937**	**33,974**	**68,911**	**31.84% (31.64–32.03%)**

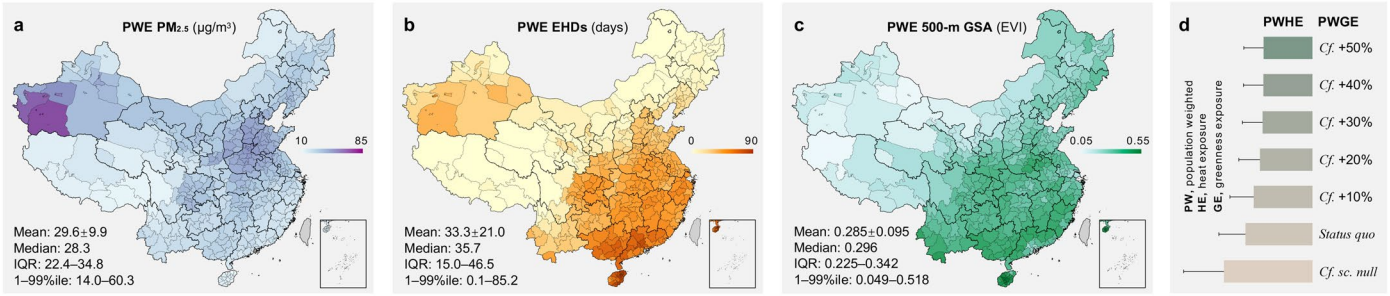
Table presents the number of pregnant participants assessed for perinatal mental health from 2013 to 2025 across 15 provinces within the ZEBRA maternity cohort, alongside the number of cases diagnosed with PMDs. Prevalence was calculated as the proportion of participants with the outcome, with 95% prediction intervals (PIs) derived from the Clopper–Pearson exact method. A DerSimonian–Laird random-effects meta-analysis indicated observable heterogeneity across provinces (I^2=97.1%, Cochran's Q=486.43, two-sided p-value<0.0001).

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Extended Data Fig. 1 | Risk associations between ambient environmental exposures and subcategories of PMDs. a–c, Associations between occurrence risks of prenatal anxiety symptoms and residential environmental exposures for (**a**) ambient $PM_{2.5}$, (**b**) EHDs, and (**c**) GSA within a 500-meter residential buffer zone. **d–f**, Associations between residential environmental exposures and occurrence risks of prenatal depressive symptoms. Seven critical periconceptional exposure windows were identified, denoted by subscripted period labels and superscripted duration in months: P_1 (preconception 0–3 months), P_2 (preconception 3–6 months), T_1 (first trimester), T_2 (second trimester), P^6 (preconception 6 months), G^6 (gestational 6 months), and P^{12} (entire periconceptional 12 months). All statistical inferences were based on records from N = 216,442 cohort participants. Open circles in forest plots indicate the central mean estimates of RRs, and whiskers indicate the 95% CIs.

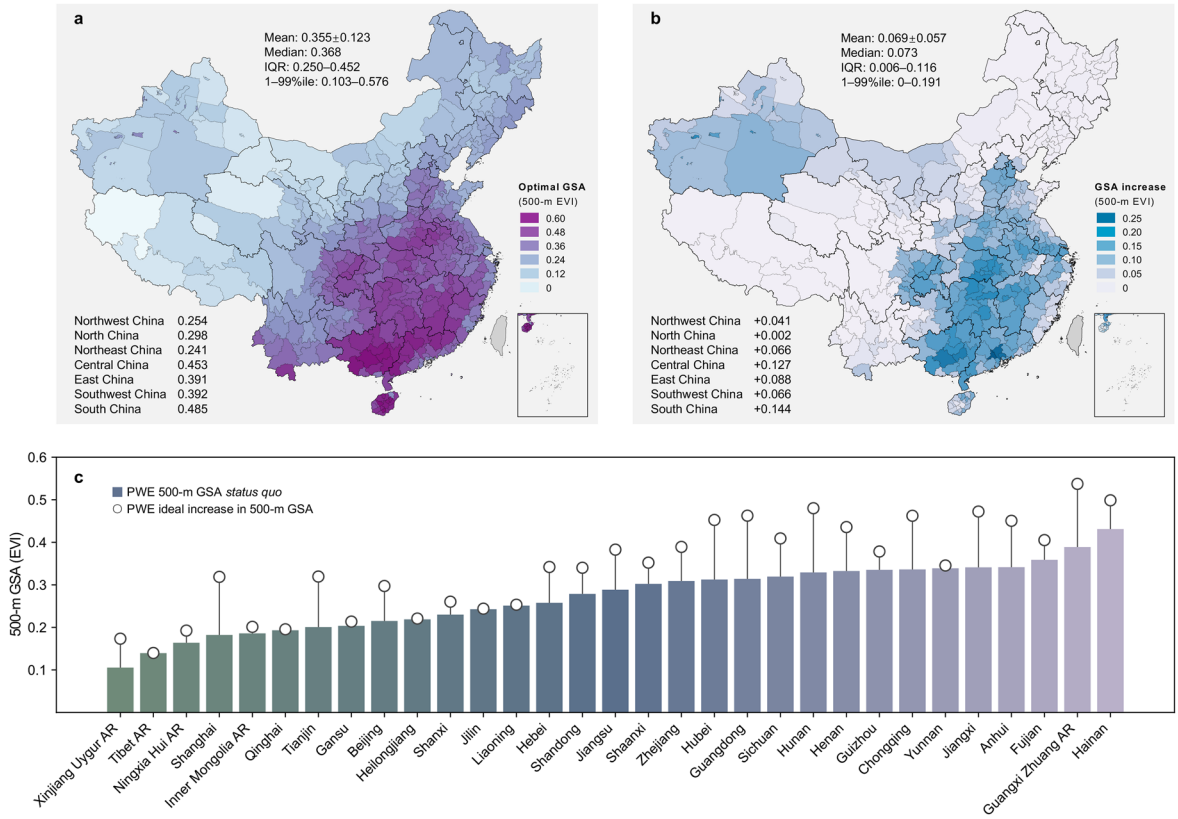
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Pathway	Description	Cases prevented (95% UI)	% Contribution
I	Heatwave reduction by urban cooling	267,900 (231,700–304,600)	26.6%
II	Buffering $PM_{2.5}$ exposure-associated risks	326,300 (200,100–463,800)	32.4%
III	Promoting outdoor physical activity	129,900 (75,900–193,700)	12.9%
IV	Direct psychological and unspecified routes	283,100 (177,400–393,300)	28.1%
TE	Total effect of "+0.1 of EVI scenario"	1,007,200 (721,900–1,355,800)	100%

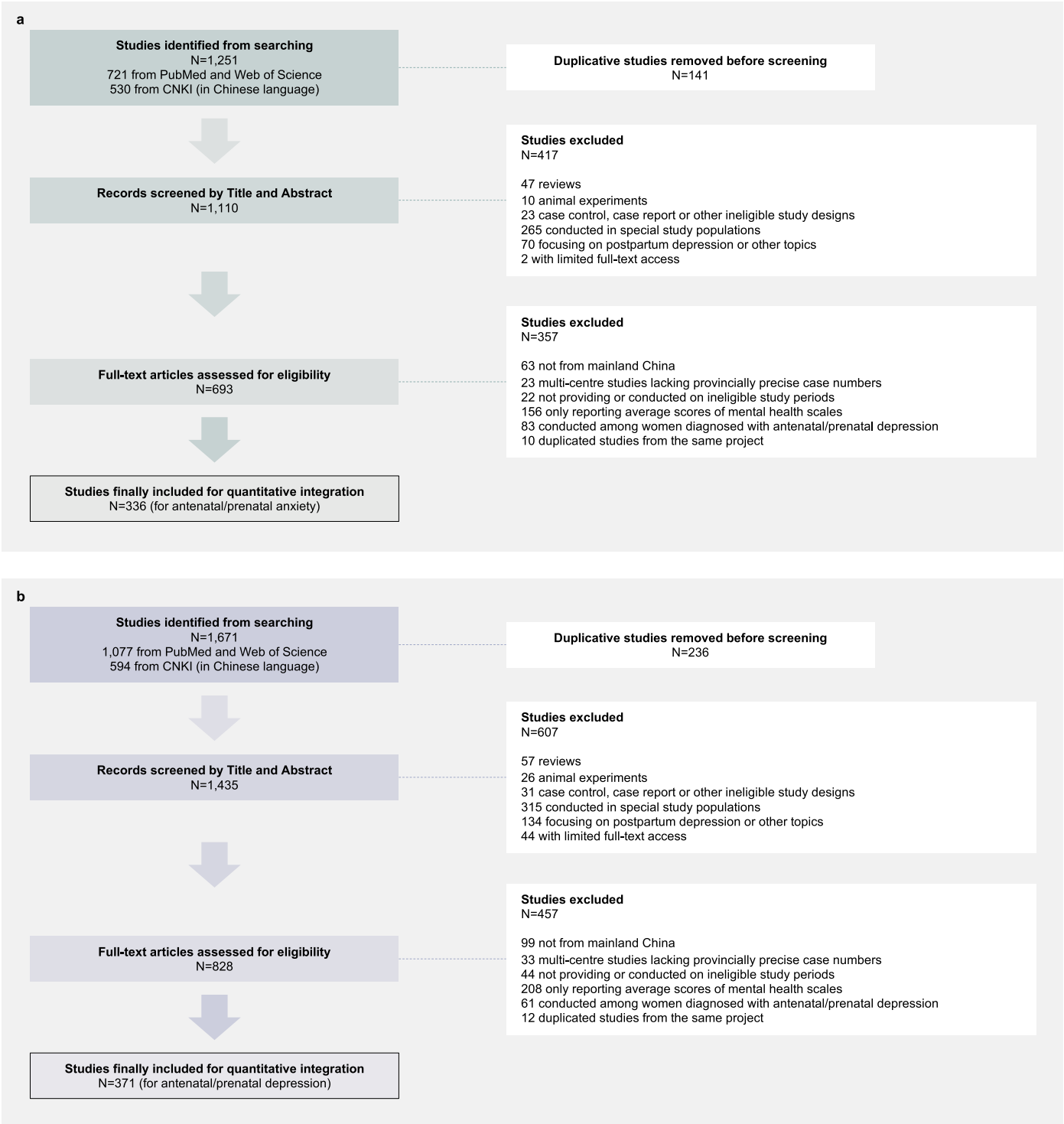
Extended Data Fig. 2 | City-level residential environmental exposure patterns across China and the mitigating effect of greenery expansion on extreme heat days. a–**c**, Population-weighted average residential ambient exposure levels across 368 Chinese prefecture-level cities, excluding Hong Kong Special Administrative Region (HKSAR), Macao SAR, and Taiwan Province, of (**a**) $PM_{2.5}$ pollution, (**b**) EHDs, and (**c**) GSA within a 500-m radius of residential locations. **d**, Responses of nationwide average exposures to EHDs against varying counterfactual levels of ambient GSA. *Cf. sc.*, counterfactual scenario; *null*, no green space coverage; *status quo*, current-state green space coverage. Bar lengths represent the nationwide mean population-weighted heat exposure (PWHE) across all high-resolution spatial grids, with the upper ends of the whiskers indicating the third quartile (75th percentile) of the nationwide grid-level geographical distribution. **e**, Increase in average EHD exposure under a hypothetical scenario of zero green space coverage. **f**, Reduction in average EHD exposure under a hypothetical 10% proportional increase in residential ambient GSA. The final table presents, in a stepwise manner, the multiple-pathway decomposition of mental health benefits under a counterfactual +0.1 EVI expansion scenario. Basemap adapted from ref. 35 under a Creative Commons license CC BY 4.0.

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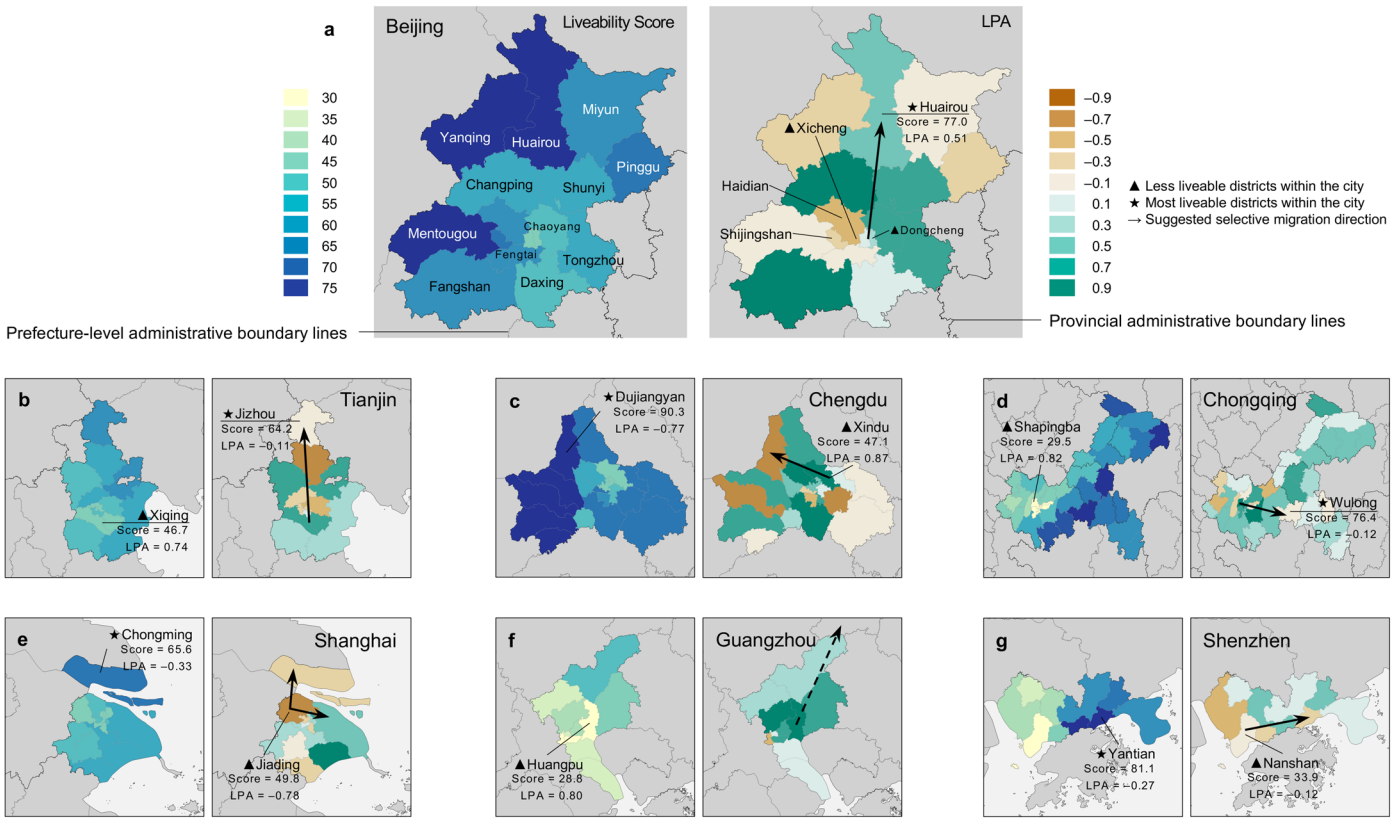
Extended Data Fig. 3 | Target optimal GSA levels required to counteract maternal PMDs associated with adverse ambient environmental exposures. **a**–**b**, National population-weighted exposure (PWE) levels of the target optimal GSA (**a**) and increases compared to current status (**b**) in the metric of the EVI, inserted with mean values for seven Chinese major geographical administrative regions. **c**, Present *status quo* of average provincial PWE-based GSA levels are shown with bars, with required increases to meet target optimal levels to counterbalance the burdens attributable to adverse ambient environmental exposures visualized in lollipop plots. Basemap adapted from ref. 35 under a Creative Commons license CC BY 4.0.

https://doi.org/10.1038/s44284-026-00534-9



Extended Data Fig. 4 | Systematic literature review for provincial-level prevalence estimates of PMDs in China. Two English-language databases (PubMed and Web of Science) and one Chinese-language database (CNKI) were systematically searched to identify epidemiological studies and case reports on (**a**) prenatal anxiety and (**b**) prenatal depression among pregnant women across various regions in China, aiming at compiling the most representative and comprehensive archives to inform provincial prevalence estimates of PMDs. Flowcharts depict the number of relevant studies identified, screened, included, and excluded at each stage of the review, with exclusion criteria transparently indicated to ensure replicability of the review process.

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Extended Data Fig. 5 | Case studies of intra-city relocation to enhance mental liveability equity in seven Chinese megacities. The figure presents a detailed case-by-case analysis of seven Chinese megacities with permanent populations exceeding 10 million: **a**, Beijing; **b**, Tianjin; **c**, Chengdu; **d**, Chongqing; **e**, Shanghai; **f**, Guangzhou; and **g**, Shenzhen. For each city, we visualize the district/county-level maternal mental health liveability scores (left panels) and the corresponding spatial concordance between population distribution and liveability (LPA; right panels). Due to page constraints, for the six cities other than Beijing, only the most and least liveable districts/counties are annotated, and illustrative directional recommendations for mental health-oriented intra-city relocation are provided based on the rankings. In Shanghai, although Chongming District exhibits the most favorable environmental conditions, limited transport accessibility to this area diminishes its practical feasibility as a relocation target. As such, Pudong New District is proposed as a viable alternative. In Guangzhou, where the overall urban mental liveability is below national average, inter-city relocation to more liveable neighboring areas such as Shaoguan is suggested (indicated with dashed arrows). Importantly, this recommendation does not constitute a critique of Guangzhou's environmental liveability. Structural challenges such as legacy air pollution in Huangpu District, insufficient green coverage in certain areas, and extreme summer humidity are real, but the local government is actively addressing these issues. It would be reductive to dismiss the liveability of Huangpu or Guangzhou based on these aspects alone, as it would overlook substantial achievements in air quality improvement, ecological richness, high-standard new district planning, and balanced development between natural environment and urban vitality. It is therefore highly anticipated that the overall environmental liveability of Guangzhou will see marked improvement in the coming years. Basemap adapted from ref. 35 under a Creative Commons license CC BY 4.0.

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Reporting Summary

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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☐	☒	A description of all covariates tested
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☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	All participant recruitment, clinical diagnosis, information registration, and invited face-to-face interviews were conducted by the ZEBRA Collaborative Network between 2013 and 2025. Individual-level participant information was manually recorded by well-trained medical staff and electronic medical record management personnel, and no automated code-based data extraction was used for these clinical or questionnaire data. Throughout the study, environmental exposure datasets were obtained by downloading data from publicly accessible websites associated with published sources, rather than through programmatic or code-based data retrieval.
Data analysis	Spatial mapping of individual-level environmental exposure levels, health impact analyses, and integrated city-level environmental assessments were conducted using Python. Epidemiological and biostatistical inference was performed using Stata/MP (version 18). Interview audio recordings were transcribed with the smartphone-embedded AI transcription tool. Subsequent integration, synthesis, and summarisation of textual information were conducted using the DeepSeek large language model. Cartographic visualisation was produced using a combination of Python and QGIS. Analytical codes directly associated with cohort participant information are available from the corresponding author(s) upon reasonable request, and codes for macro-scale health impact analyses are publicly accessible at https://github.com/csuen27/zebra-mental-liveability.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
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For urban residential environmental exposure assessment, daily high-spatial-resolution ambient $PM_{2.5}$ and O_3 concentrations can be obtained from the Tracking Air Pollution (TAP) database (http://tapdata.org.cn), which integrates ground-based observations, satellite remote sensing, chemical transport model simulations, and land cover information using an ensemble machine learning framework to generate daily exposure estimates at a 1×1 km² spatial resolution. Ambient NO_2 concentrations are provided by the China High-resolution Air Pollution (CHAP) database. Ambient temperature and humidity archives can be retrieved from the ECMWF ERA5 reanalysis products (https://www.ecmwf.int/en/forecasts/datasets) at a 0.1°×0.1° spatial resolution. Green space accessibility (GSA) was assessed using the Enhanced Vegetation Index (EVI) from the MODIS Vegetation Index product (https://modis.gsfc.nasa.gov/data/dataprod), resampled to a 0.5×0.5 km² grid and aggregated within 1–3 km residential buffers at 16-day temporal intervals. The geographical distribution of municipal functional facilities can be supported by the Essential Urban Land Use Categories Map across China (EULUC China 2.0). Spatially resolved population density can be sourced from the LandScan population data platform (https://landscan.ornl.gov).

Individual-level records of ZEBRA maternity cohort participants, including socioeconomic characteristics, residential addresses, medical histories and current pregnancy-related health status, were collected across multiple cities in China. Owing to ethical and legal restrictions and the need to protect participant privacy, these individual-level data cannot directly be available to the public. However, an appropriately anonymised and de-identified version of the dataset, limited to the variables necessary to reproduce and verify the analyses reported in the present study, has been deposited (https://github.com/csuen27/zebra-maternity-cohort). The repository will be actively maintained and updated in response to research-based requests for 12 months following publication. Researchers seeking further access to the ZEBRA maternity cohort should submit a brief research proposal to the principal investigators and corresponding authors, A/Prof Haitong Zhe Sun or Prof He-Feng Huang, for review. Requests will usually receive a response within four weeks. Following approval by the ZEBRA Collaborative Committee and the Health Commission of Zhejiang Province, restricted access can be granted for research collaboration purposes for reasons of participant confidentiality and data-sharing agreements with the Chinese Medical Association (CMA).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	This study focused on maternal health during pregnancy. All participants were individuals assigned female sex at birth who were pregnant at the time of recruitment. Sex was recorded based on clinical and medical records. Gender identity was not separately assessed in this study.
Reporting on race, ethnicity, or other socially relevant groupings	Information on age at pregnancy, educational attainment, and urban or rural residence was obtained from nationally standardised medical and health insurance registration records. Occupation, household income level (categorised as high, moderate, or low), household solid fuel use (yes/no), smoking or second-hand smoke exposure status (yes/no), preconception planning intention, pet companionship (dog, cat, or none), and moderate-intensity physical activity (yes/no) were collected through structured, interviewer-guided questionnaires administered by trained healthcare professionals. Use of assisted reproductive technology (yes/no) and gestational body mass index (BMI) change were obtained from routine hospital clinical records during pregnancy. Race and ethnicity were not separately collected or analysed in this study, as these variables are not routinely recorded in the underlying medical registration systems and were not central to the study objectives. Control of confounding effects was primarily guided by Directed Acyclic Graphs (DAGs), constructed to avoid inappropriate adjustment for potential mediators and colliders, and implemented using DAGitty software (https://dagitty.net).
Population characteristics	Among the 216,442 pregnant women included in the present analysis, 31.8% were diagnosed with antenatal mental distress symptoms (AMDs). The mean age at pregnancy was 31.2 years, with 4.0% aged <25 years, 34.3% aged 25–29 years, 40.2% aged 30–34 years, and 21.5% aged ≥35 years. With respect to highest educational attainment, 12.4% of participants had a high school education or below, 33.8% had a college-level education, 36.4% held an undergraduate degree, and 17.3% had a graduate degree or above.
Recruitment	Participants were recruited as part of the ongoing ZEBRA maternity cohort through collaborating hospitals across multiple provinces in China. Eligible pregnant women were approached in person during routine antenatal visits by trained clinical staff at participating hospitals and invited to take part in the cohort. Recruitment was conducted consecutively according to predefined inclusion criteria, with informed consent obtained prior to data collection. Because enrolment was embedded within routine clinical care and followed a comprehensive "register-at-visit" procedure, participation did not rely on self-referral or targeted advertising. Refusal to participate occurred passively during routine care encounters and was not linked to specific exposure or outcome profiles, making systematic selection bias unlikely. The cohort covers 15 provinces spanning across China, providing broad geographic and environmental diversity. Representativeness with respect to environmental exposures was further supported by comparisons between region-specific exposure distributions in the cohort and corresponding population-level exposure profiles across China, as described in the Methods.
Ethics oversight	The study was approved by the Ethics Committee of the Women's Hospital, Zhejiang University School of Medicine

Ethics oversight	(IRB-20220189-R, IRB-20240370-R), and the Institutional Review Board of International Peace Maternity and Child Health Hospital (GKLW2019-11). Written informed consent was obtained from all participants.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

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Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This study is a large-scale, multi-centre observational cohort analysis based on the ZEBRA maternity cohort, comprising 216,442 pregnant women recruited across 15 Chinese provinces between 2013 and 2025. The study examined associations between antenatal mental distress symptoms (AMDs) and a wide range of risk determinants, including sociobehavioural characteristics and residential environmental exposures, with assessment of exposure–response relationships, interactions between exposures, and effect modification by maternal characteristics. Analyses were conducted at the individual epidemiological level and synthesised to the population level, with a specific focus on city-scale quantification of mental liveability. No experimental manipulation or randomisation was performed.
Research sample	The research sample (individual-level participants) consisted of 216,442 enrolled pregnant women selected from an initial cohort of 257,255 participants after exclusion of individuals with missing key information. Participants were recruited through routine antenatal care at collaborating hospitals and represent a broad hospital-based pregnant population with diverse socioeconomic, behavioural, medical, and environmental exposure profiles. Participants were chosen to enable individual-level assessment of antenatal mental health in relation to residential environmental exposures, while allowing aggregation to population and city scales for comparative urban analyses. The study population was restricted to women with complete mental health assessments and valid residential information during pregnancy. No experimental manipulation was applied to cohort participants.
Sampling strategy	No formal sampling procedure was applied in this study. Participants were enrolled consecutively at collaborating hospital sites located in selected cities across 15 Chinese provinces, following a "register-at-visit" approach whereby all eligible pregnant women attending routine antenatal care were invited to participate. All participants meeting predefined data quality criteria were included in the final epidemiological analyses.
Data collection	All participant recruitment, clinical diagnosis, information registration, and invited face-to-face interviews were conducted by the ZEBRA Collaborative Network between 2013 and 2025. Residential environmental exposure tracking for each participant—including $PM_{2.5}$, extreme temperatures, green space accessibility (GSA), O_3, and NO_2—was performed through secondary analyses of previously published and publicly available environmental databases. Most environmental tracking databases provided coverage before 2025; thus, for a small proportion of cohort participants enrolled in 2025, residential environmental exposures for year 2025 were approximated using corresponding calendar-day values from the nearest available year, given the strong seasonal and cyclical characteristics of these environmental variables.
Timing and spatial scale	The study analysed pregnant women who delivered between January 2013 and June 2025, covering 15 provinces and 27 prefecture-level cities in China. Residential environmental exposures were tracked at a daily temporal resolution.
Data exclusions	A total of 19,039 participants without mental health assessments were excluded. In addition, 20,197 participants with more than 20% missing data or unresolvable data errors were excluded. Sixty-four participants with gestational ages below 20 weeks or above 44 weeks, and 232 participants with maternal age <18 or >45 years at pregnancy were also excluded. These exclusion criteria were applied to ensure data validity and analytic consistency and to minimise the influence of implausible or unverifiable records. All remaining participants met predefined quality control criteria and were included in the final epidemiological analyses.
Reproducibility	This study used publicly accessible, peer-reviewed environmental exposure databases that have been widely adopted in the literature. The core analytical approaches followed established methods described in highly cited previous studies. To facilitate reproducibility, the primary analytical code will be made publicly available on GitHub upon publication.
Randomization	Randomisation was not applicable throughout the present study.
Blinding	Blinding was not applicable in the present study. The research involved face-to-face interviews with participants, questionnaire-based assessment of mental health status, and direct information exchange between healthcare staff and participants during routine clinical care. As such, neither participants nor investigators were blinded to exposure or outcome status.

Did the study involve field work? ☐ Yes ☒ No

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☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
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